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Silicon to Software™

Chip Finish & STAR RC

Synopsys University Program

WEEK 14 Chip Finish & STAR RC

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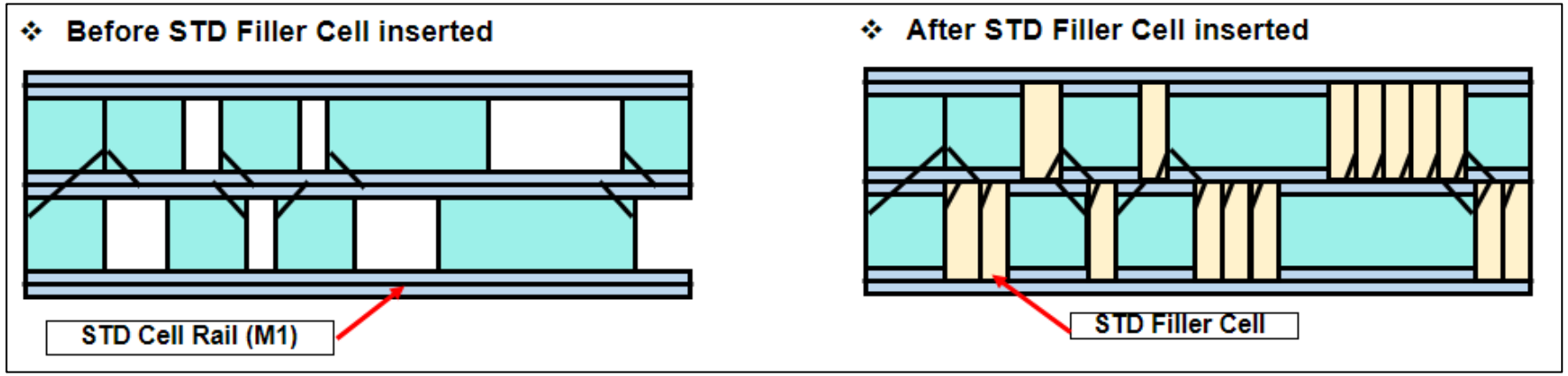
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WEEK 15 : Introduction to Chip Finish

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Introduction to Chip Finish : STD Filler Cell

- **STD Filler Cell insertion**

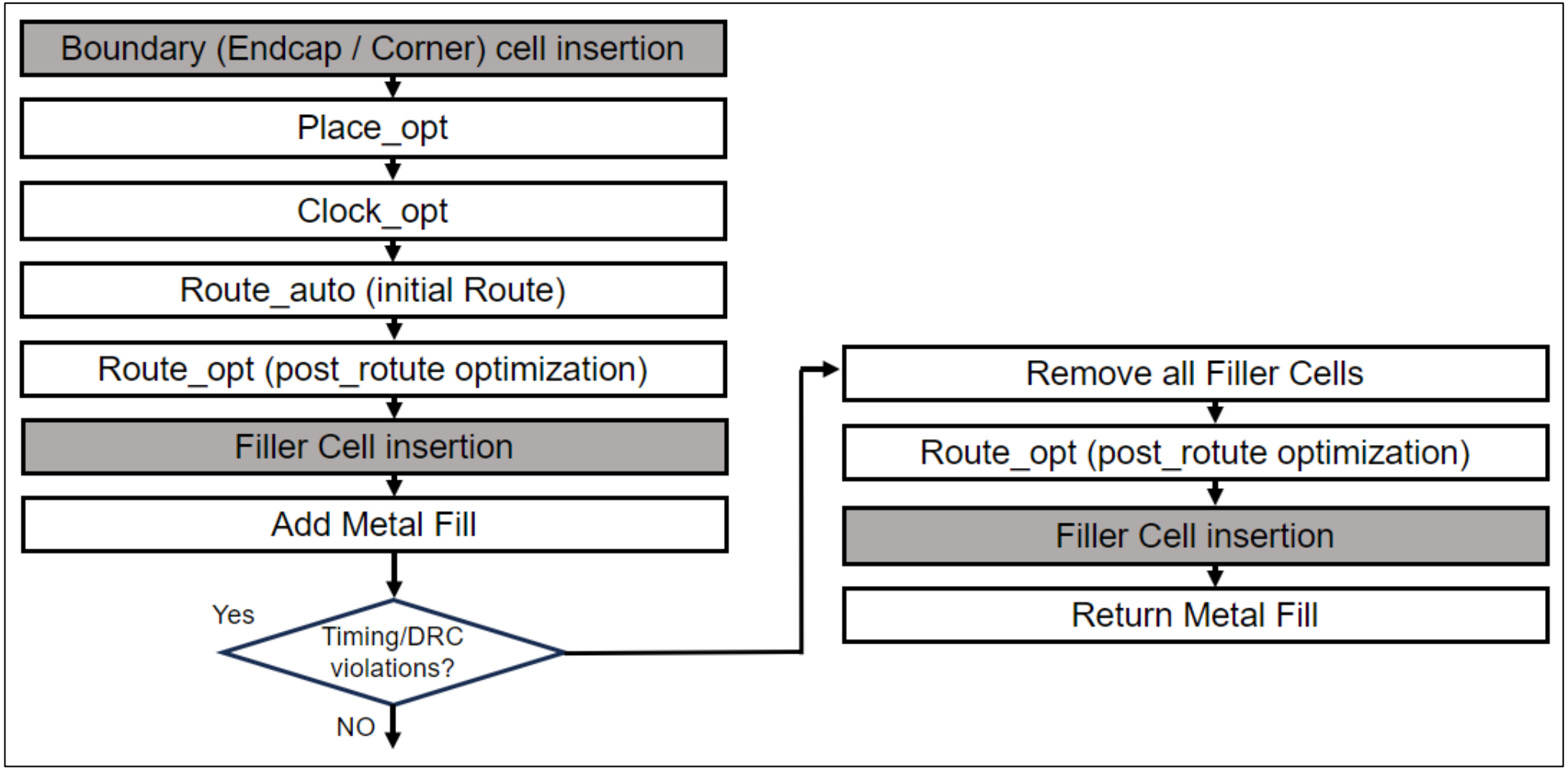


- STD Filler Cells can be inserted before or after Detailed Routing (clock, signal, routing)
- If Fillers contain metal routing other than Pre-Routing then Fillers should be inserted before Routing
- Width of the smallest Filler Cell is the Placement Grid Width
- Once Fillers are inserted then the placement is fixed and tool can’t move Cells for further optimization
- PG connection and checking are done afterwards in separate steps
- May have to iterate with different sized fillers

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Introduction to Chip Finish : STD Filler Cell

- **STD Filler Cell insertion**



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Introduction to Chip Finish : STD Filler Cell

- **STD Filler Cell insertion**

```
####################
#insert_filler
####################

set SH_FILLERS "*/SHFILL128_HVT */SHFILL64_HVT */SHFILL3_HVT */SHFILL2_HVT */SHFILL1_HVT"
create_stdcell_fillers -lib_cells $SH_FILLERS -rules {post_route_auto_delete} ;# it allows route_opt & eco_opt to automatica
lly remove filler cells
```

  - Connect pg net

```
#connect pg_net for all STD filler cell#
set FILLERS [get_cells -filter "full_name =~ *SHFILL*"]

set i 0

foreach x $FILLERS {
set pins [get_pins -of_objects $x -filter "port_type == power"]
connect_pg_net -net VDD [get_pins $pins]

incr i
}

foreach x $FILLERS {
set pins [get_pins -of_objects $x -filter "port_type == ground"]
connect_pg_net -net VSS [get_pins $pins]

incr i
}
```

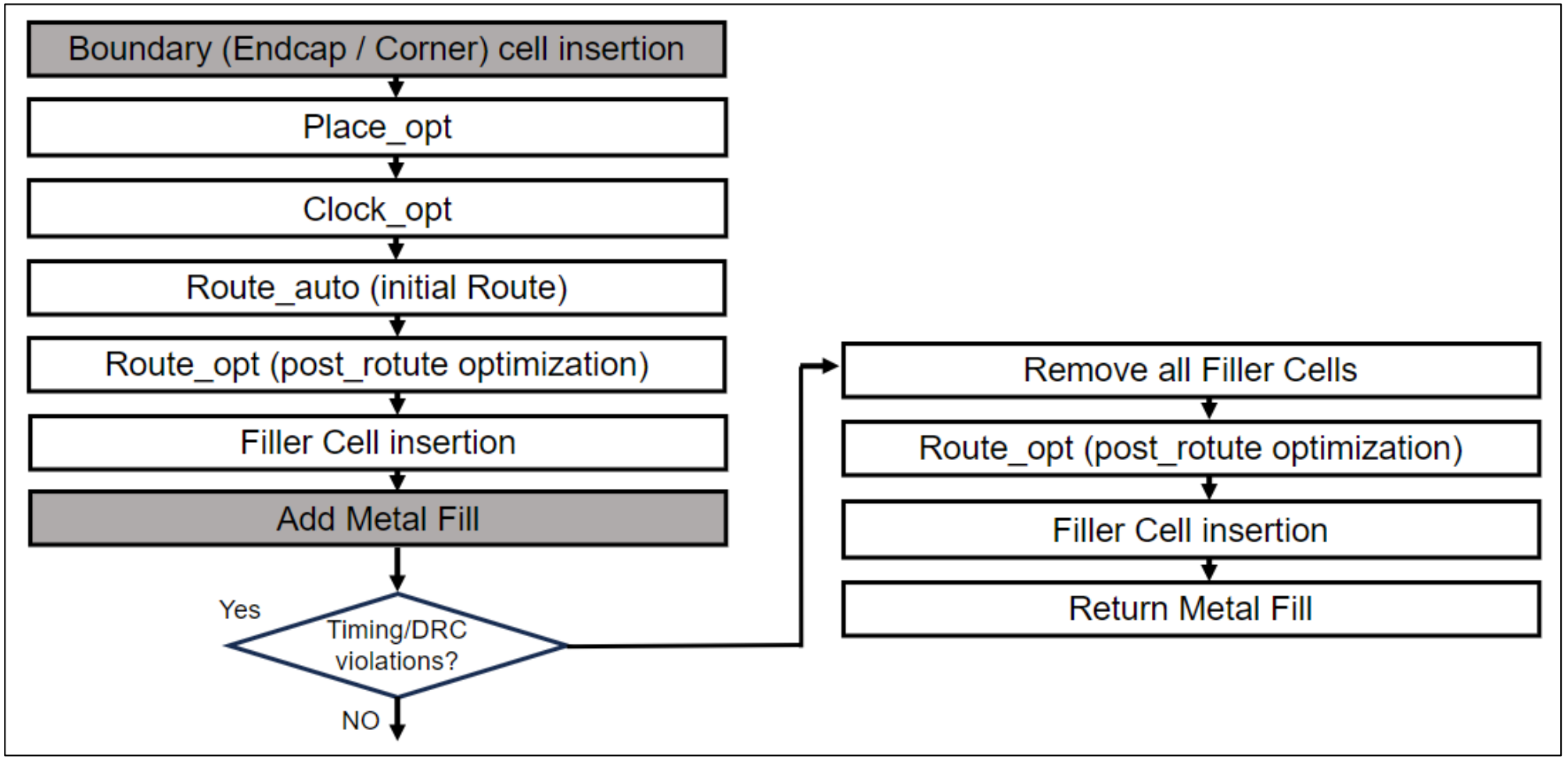
  - Remove Filler cells with violations

```
remove_stdcell_fillers_with_violation
```


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Introduction to Chip Finish : STD Filler Cell

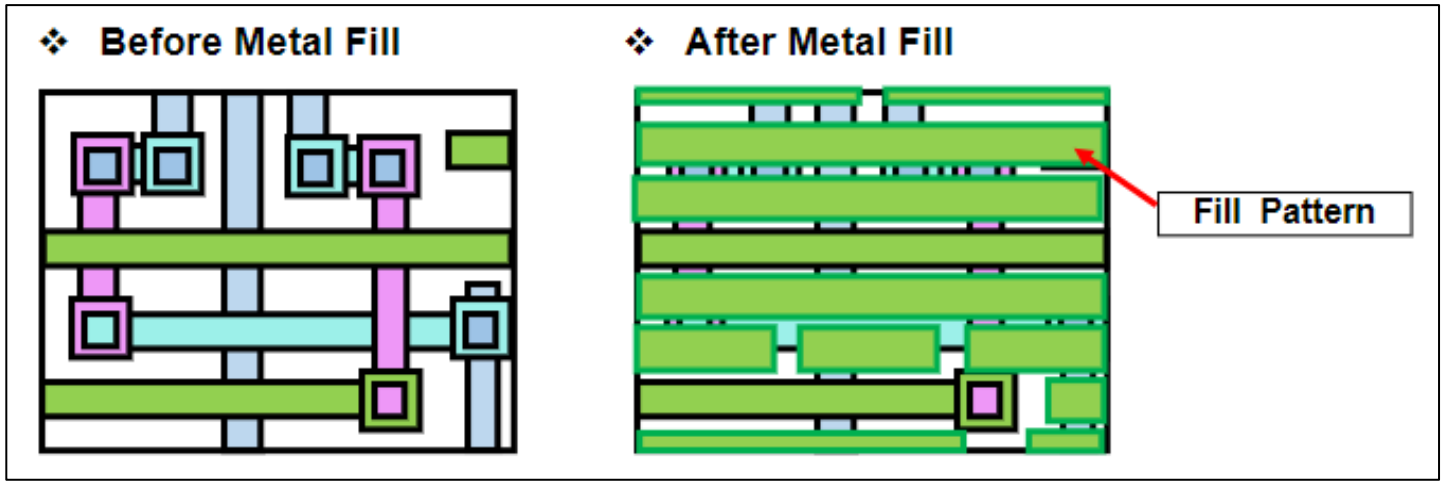
- **Add Metal Fill**



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Introduction to Chip Finish : Metal Fill

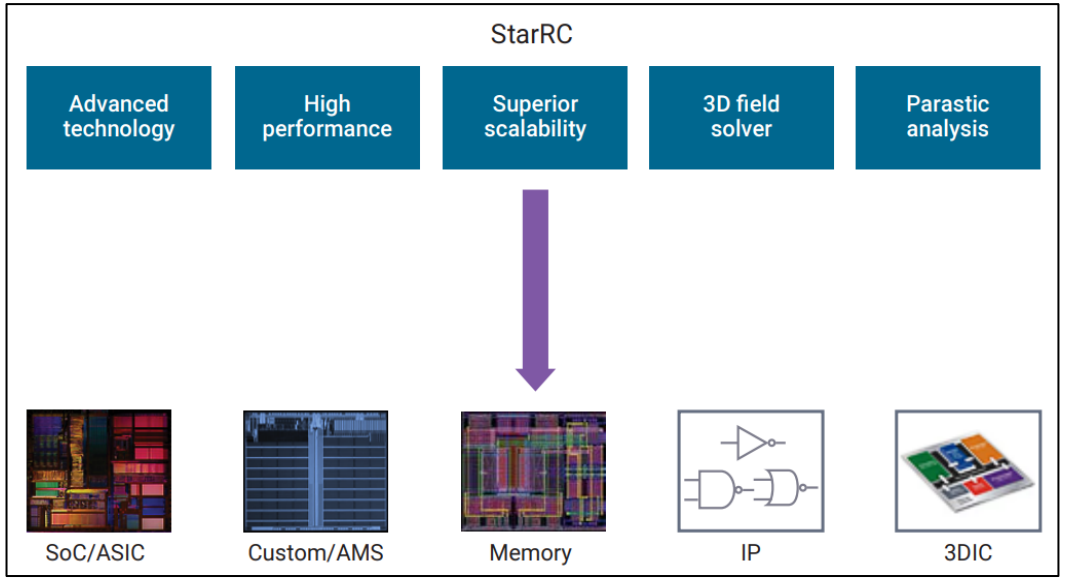
- **What is Metal Fill**
  - Whether performed by the designer or by the foundry, Metal fill is a mandatory step at advanced nodes to ensure manufacturability and high yield
  - Metal fill involves filling the empty spaces near the design with metal polygons to ensure regular planarization of the wafer (Foundry-mandated fill requirements stipulate that the fill density be within specified maximum and minimum parameters)
  - While regular planarization is crucial to prevent over or under polishing of signal routes and active circuitry, metal fill can also negatively affect timing due to increased
  - Designer must ensure that the Metal fill is done without impacting timing-critical nets
  - The fill methodology should allow easy removal and re-insertion of ill without violating timing of the nearby nets



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Introduction to Chip Finish : STAR RC

- **What is STAR RC ?**




```
                        StarRC (TM)

                Version S-2021.06-SP5-5 for linux64 - Aug 06, 2023

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```

- StarRC is the EDA industry's gold standard for parasitic extraction
- Synopsys StarRC provides a silicon-accurate and high-performance extraction solution for SoC (System on Chip), custom digital, analog/mixed-signal, memory and 3DIC designs
- The Synopsys StarRC solution offers modeling of physical effects for advanced process technologies, including FinFET technologies at 16nm, 14nm, 10nm, 7nm, 5nm, 3nm and beyond
- Its seamless integration with industry standard digital and custom implementation systems, physical verification timing, signal integrity, power, thermal and circuit simulation flows deliver unmatched ease-of-use and productivity to speed design closure and signoff verification

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Introduction to Chip Finish : STAR RC

- **Benefits of STAR RC**
  - Foundry gold standard for extraction accuracy with broadest qualification and adoption
  - Leader in advanced modeling, including FinFET and color-aware multi-patterning at 10nm/7nm/5nm/3nm and beyond
  - High performance and capacity for gate and transistor-level extraction, enabled by multi-core distributed processing and simultaneous multi-corner extraction
  - Tightly integrated with industry leading Synopsys IC Compiler II and Synopsys PrimeTime solutions for faster full - flow ECO turn-around time
  - Unified Rapid3D fast field solver for critical net, IP, and custom circuit extraction 3
  - Advanced netlist reduction features for faster simulation turn-around time
  - Inductance extraction for high frequency digital RLC clock net analysis
  - 3DIC extraction solution for interposer and stacked die technologies
  - Integration with Synopsys IC Validator physical verification solution, Synopsys PrimeSim circuit simulation technologies, Synopsys Custom Compiler design environment and other third-party implementation and custom design solutions for increased designer productivity

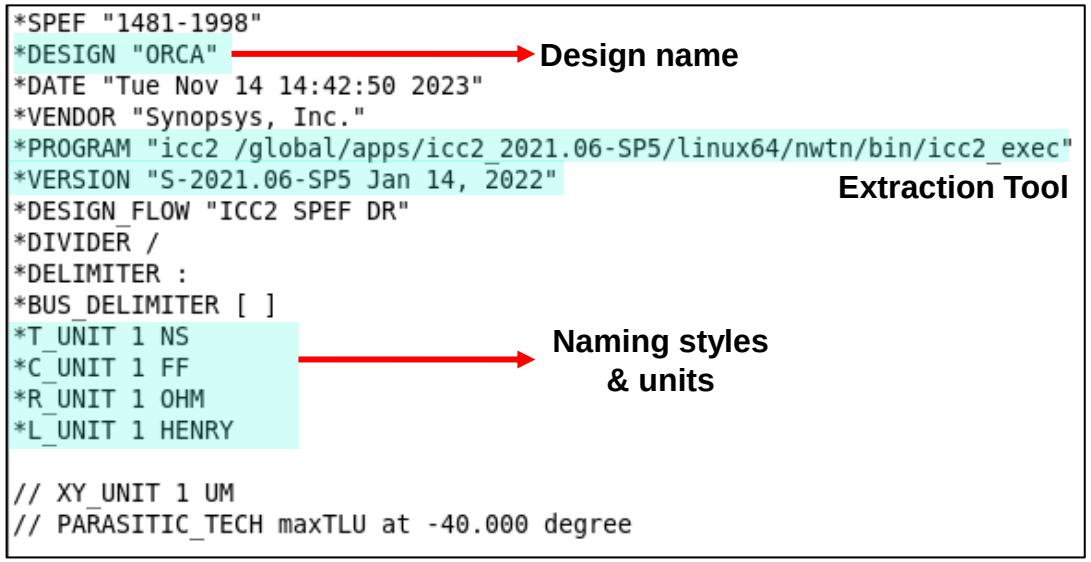
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Introduction to Chip Finish : STAR RC

- **Extraction .spef file use STAR RC**
  - **What is .spef file?**
    - SPEF stands for Standard Parasitic Exchange Format
    - .spef file is a file format to save parasitic information extracted by the P&R tool in VLSI design
    - .spef file has four main sections (a header section, a name map section, a top level port section, the main parasitic description section)
  - **Structure of .spef file**
    - **Header Information**

      The header section containing information about

      1) Design name
      2) Parasitic extraction tool
      3) Naming styles
      4) Units



❖ When reading .spef file, it is important to check the header for units as they vary Across tools (By default, spef from Astro will be in pF and kOhm while spef from Star-RCXT will be in fF and Ohm)

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Introduction to Chip Finish : STAR RC

- **Extraction .spef file use STAR RC**
  - **Port section**

    The port section is simply a list of the top level ports in a design (They are annotated as input, output or bidirect with an I, O or B)

```
*NAME_MAP
*1 scan_en
*2 test_mode
*3 power_save
*4 sys_clk
```

  - **Parasitics**
    - Each extracted net will have a *D_NET section
    - This will usually consist of a *D_NET line, a *CONN section, a *CAP section, *RES section and a *END line
    - Single pin nets will not have a *RES section
    - Nets connected by abutting pins will not have a *CAP section

    ❖ **D_NET Line**

```
*D_NET *13245 0.4618512 //LENGTH 7.171 LUMPCC 0 DR
```

    - ***D_NET line tells the net name and the net's total capacitance**
    - **This capacitance will be the sum of all the capacitances in the *CAP section**

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Introduction to Chip Finish : STAR RC

- **Extraction .spef file use STAR RC**
  - **Parasitics**
    - ❖ ***CONN Section**

```
*CONN
*I *8949:Y O *L 0 *C 928.164 704.724
*I *223:FB_CLK I *L 0.341195 *C 927.078 709.938
*N *13245:2 *C 927.982 705.180
*N *13245:3 *C 928.164 704.998
*N *13245:4 *C 927.078 706.580
*N *13245:5 *C 927.252 706.580
*N *13245:6 *C 927.252 705.193
*N *13245:7 *C 927.265 705.180
*N *13245:8 *C 928.150 705.180
*N *13245:9 *C 928.164 705.167
*N *13245:10 *C 928.151 705.180
*N *13245:11 *C 928.164 705.166
*N *13245:12 *C 928.164 704.737
```

- **The *CONN section lists the pins connected to the net**
- **A connection to a cell instance starts with a *I**
- **A connection to a top level ports starts with a *P**
- **Syntax of the *CONN entries is :**

  I <pin name> <direction> *C <x y coordinate> <loading or driving information>

- **The pin name is the name of the pin**
- **The direction will be I, O or B (input, output or Bidirect)**
- **The xy coordinate will be the location of the pin in the layout**
- **For an input, the loading information will be *L and the pin's capacitance**
- **For an output, the driving information will be *D and the driving cell's type**
- **Coordinates for *P port entries may not be accurate because some extraction tools look for the physical location of the logical port rather then the location of the corresponding pin**

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Introduction to Chip Finish : STAR RC

- **Extraction .spef file use STAR RC**
  - **Parasitics**
    - ❖ ***CAP Section**

```
*CAP
0 *8949:Y 0.001
1 *223:FB_CLK 0.114599
2 *13245:2 0.008580254
3 *13245:3 0.008161967
4 *13245:4 0.1220509
5 *13245:5 0.06014389
6 *13245:6 0.05091337
7 *13245:7 0.02820281
8 *13245:8 0.02820263
9 *13245:9 0.004604246
10 *13245:10 0.005022287
11 *13245:11 0.01575668
12 *13245:12 0.01461314
```

- **The *CAP section provides detailed capacitance information for the net**
- **Entries in the *CAP section come in two forms, on for a capacitor lumped to ground and one for a coupled capacitor**
- **A capacitor lumped to ground**
  1) **Identifying integer**
  2) **Node name**
  3) **Capacitance value of this node**
- **Coupling capacitor**
  1) **Identifying integer**
  2) **Two nodes name**
  3) **The values of the coupling capacitor between these two nodes**

  ❖ ***If net1 is coupled to net2, the coupling capacitor will be listed in each net's *CAP section***

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Introduction to Chip Finish : STAR RC

- **Extraction .spef file use STAR RC**
  - **Parasitics**
    - ❖ ***RES & END Section**

```
*RES
0 *8949:Y *13245:12 1
1 *13245:7 *13245:6 0.5
2 *13245:6 *13245:5 2.476786
3 *13245:9 *13245:8 0.5
4 *13245:9 *13245:3 0.2816667
5 *13245:8 *13245:7 1.580357
6 *13245:10 *13245:9 0.5
7 *13245:10 *13245:2 0.2816667
8 *13245:11 *13245:10 0.5
9 *13245:12 *13245:11 0.7660715
10 *13245:4 *223:FB_CLK 5.996429
11 *13245:5 *13245:4 0.3107143

*END
```

- **The *RES section provides the resistance network for the net**
- **Entries in *RES section contain,**
- **The resistance network for a net can be very complex**
- **.spef file can contain resistor loops or seemingly ridiculously huge resistors even if the layout is a simple point to point route**
- **This is due how the extraction tool cuts nets into tiny pieces for extraction and then mathematically stitches them back together when writing .spef file**

  - **Parasitic Values**

❖ **Ex) I_reg_top/GRC/U9743:E 0.946054:1.03456:1.41343**

- Example show a single parasitic value for each capacitor or resistor
- It is up to the parasitic extraction and delay calculation flow to decide which corner this value represents
- SPEF also allows for (min | typ | max) values to be reported
- Some tools will report min : max pairs and it is expected that tools may report many corners in the feature

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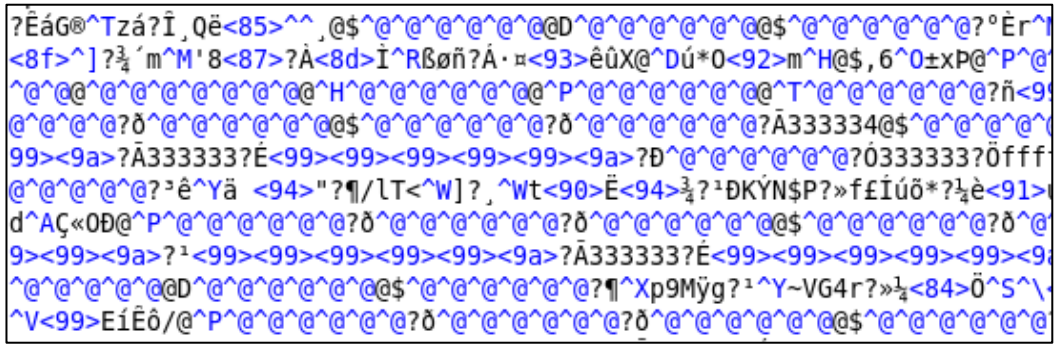
Introduction to Chip Finish : STAR RC

- **Nxtgrd file**
  - Nxtgrd file is an output file used in VLSI design for layer parasitic extraction
  - Nxtgrd file is a database containing capacitance, resistance, inductance, and layer/via information that first undergoes process characterization and is then compiled by the grdgenxo command function
  - It is used to generate parasitic formats for the extraction tools (STAR RC)
  - Nxtgrd file is generated from ITF file with grdgenxo utility in STAR RC

- **Structure of Nxtgrd file**

```
CONDUCTOR M5 { THICKNESS = 0.078 WMIN=0.056 SMIN=0.056 RPSQ=0.1 }
DIELECTRIC D5 { THICKNESS = 0.72 ER=3.9 }
CONDUCTOR M4 { THICKNESS = 0.078 WMIN=0.056 SMIN=0.056 RPSQ=0.1 }
DIELECTRIC D4 { THICKNESS = 0.72 ER=3.9 }
CONDUCTOR M3 { THICKNESS = 0.078 WMIN=0.056 SMIN=0.056 RPSQ=0.1 }
DIELECTRIC D3 { THICKNESS = 0.72 ER=3.9 }
CONDUCTOR M2 { THICKNESS = 0.078 WMIN=0.056 SMIN=0.056 RPSQ=0.1 }
DIELECTRIC D2 { THICKNESS = 0.72 ER=3.9 }
CONDUCTOR M1 { THICKNESS = 0.078 WMIN=0.05 SMIN=0.05 RPSQ=0.1 }
DIELECTRIC D1 { THICKNESS = 0.3 ER=3.9 }
```

- **Nxtgrd file is a file generated on the basis of ITF file**



- **Nxtgrd file cannot be opened with gvim to view to contents**

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Introduction to Chip Finish : STAR RC

- **ITF file**
  - ITF stands for Interconnect Technology File
  - ITF file contains a description of the process cross-section and connectivity section
  - ITF file describes the thicknesses and physical attributes of the conductor and dielectric layers
  - ITF file used to extract RC value for the chip (STAR RC)
  - ITF file also used to generate TLU/TLUplus or Nxtgrd files
  - ITF file contains statements to describe conductor layers, dielectric layers, via layers and etc.

- **Structure of ITF file**

```
TECHNOLOGY=saed32nm_1p9m_Cmin
DIELECTRIC PASS1 { THICKNESS = 3.00 ER=3.9 }
```

- **TECHONOLGY : value**

- **CONDUCTOR statement**

```
CONDUCTOR M2 { THICKNESS = 0.078 WMIN=0.056 SMIN=0.056 RPSQ=0.1 }
DIELECTRIC D2 { THICKNESS = 0.72 ER=3.9 }

CONDUCTOR M1 { THICKNESS = 0.078 WMIN=0.05 SMIN=0.05 RPSQ=0.1 }
DIELECTRIC D1 { THICKNESS = 0.3 ER=3.9 }
```

- **THICKNESS : measured from top of the dielectric layer below it**
- **WMIN : Minimum width of the conductor**
- **SMIN : Minimum spacing between two conductors**
- **RPSQ: sheet resistance, units: ohms/sq**

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Introduction to Chip Finish : STAR RC

- **ITF file**
  - **Structure of ITF file**
    - **DIELECTRIC statement**

```
CONDUCTOR M2 { THICKNESS = 0.078 WMIN=0.056 SMIN=0.056 RPSQ=0.1 }
DIELECTRIC D2 { THICKNESS = 0.72 ER=3.9 }

CONDUCTOR M1 { THICKNESS = 0.078 WMIN=0.05 SMIN=0.05 RPSQ=0.1 }
DIELECTRIC D1 { THICKNESS = 0.3 ER=3.9 }
```

- **THICKNESS : measured from top of the dielectric layer below it by default**
- **ER : Dielectric constant of the layer**

    - **VIA statement**

```
VIA SUBCONT  { FROM=SUBSTRATE TO=DIFF RPV=0.00001 AREA=0.0144}
VIA DIFFCONT { FROM=DIFF TO=M1 AREA=0.001764 RPV=150 }
VIA POLYCONT { FROM=POLY TO=M1 AREA=0.001764 RPV=135 }
```

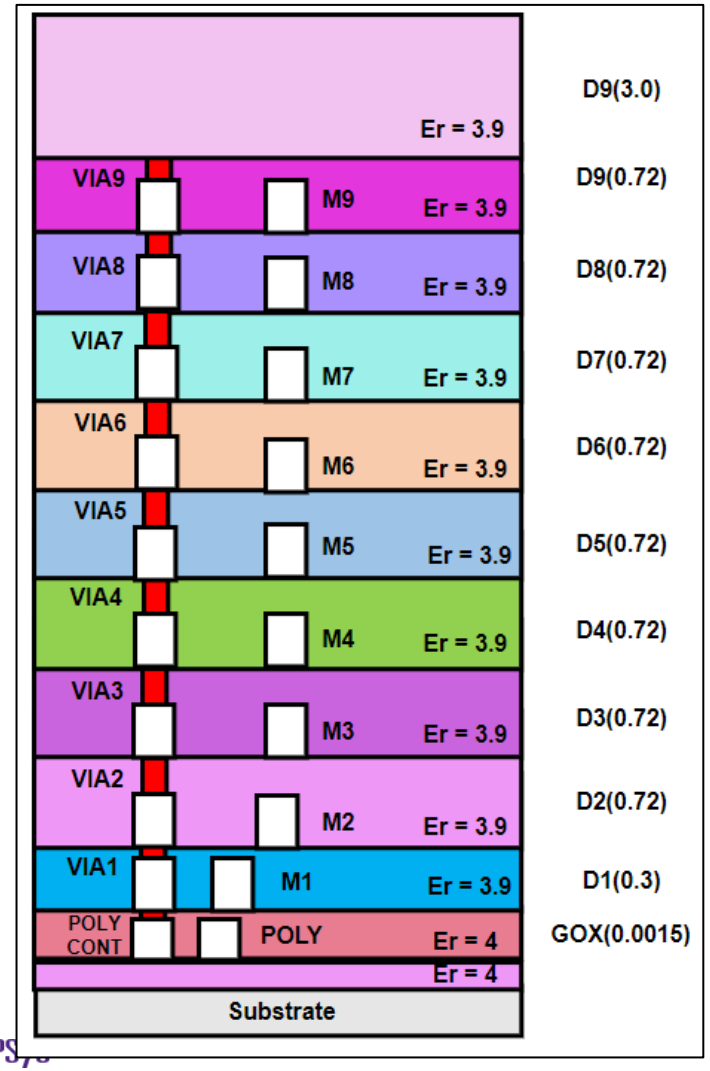
- **RPV : resistance per default via, units (ohms)**
- **AREA : Area of defualt via, units (square-micorns)**

  - **ITF file contain data**
    - **Conformal dielectrics (MEASURED_FROM = TOP_OF_CHIP)**
    - **Emulation metal fill**
    - **Drop factor (DROP_FACTOR)**
    - **Etch (ETCH)**
    - **Width & spacing based etch**
    - **Density based thickness**
    - **Width dependent RPSQ**
    - **Temperature derating**

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Introduction to Chip Finish : STAR RC

- **ITF file**
  - **Example of ITF file**




```
CONDUCTOR MRDL { THICKNESS = 0.23 WMIN=2 SMIN=2 RPSQ=0.35 }
DIELECTRIC PASS2 { THICKNESS = 3.00 ER=3.9 }

CONDUCTOR M9 { THICKNESS = 0.158 WMIN=0.16 SMIN=0.16 RPSQ=0.28 }
DIELECTRIC D9 { THICKNESS = 0.72 ER=3.9 }

CONDUCTOR M8 { THICKNESS = 0.078 WMIN=0.056 SMIN=0.056 RPSQ=0.1 }
DIELECTRIC D8 { THICKNESS = 0.72 ER=3.9 }

CONDUCTOR M7 { THICKNESS = 0.078 WMIN=0.056 SMIN=0.056 RPSQ=0.1 }
DIELECTRIC D7 { THICKNESS = 0.72 ER=3.9 }

CONDUCTOR M6 { THICKNESS = 0.078 WMIN=0.056 SMIN=0.056 RPSQ=0.1 }
DIELECTRIC D6 { THICKNESS = 0.72 ER=3.9 }

CONDUCTOR M5 { THICKNESS = 0.078 WMIN=0.056 SMIN=0.056 RPSQ=0.1 }
DIELECTRIC D5 { THICKNESS = 0.72 ER=3.9 }

CONDUCTOR M4 { THICKNESS = 0.078 WMIN=0.056 SMIN=0.056 RPSQ=0.1 }
DIELECTRIC D4 { THICKNESS = 0.72 ER=3.9 }

CONDUCTOR M3 { THICKNESS = 0.078 WMIN=0.056 SMIN=0.056 RPSQ=0.1 }
DIELECTRIC D3 { THICKNESS = 0.72 ER=3.9 }

CONDUCTOR M2 { THICKNESS = 0.078 WMIN=0.056 SMIN=0.056 RPSQ=0.1 }
DIELECTRIC D2 { THICKNESS = 0.72 ER=3.9 }

CONDUCTOR M1 { THICKNESS = 0.078 WMIN=0.05 SMIN=0.05 RPSQ=0.1 }
DIELECTRIC D1 { THICKNESS = 0.3 ER=3.9 }

CONDUCTOR POLY { THICKNESS = 0.0066 WMIN=0.03 SMIN=0.086 RPSQ=15 }
DIELECTRIC GOX { THICKNESS = 0.0015 ER=4 }
CONDUCTOR  DIFF   {IS_DIFF THICKNESS=0.05 WMIN=0.044 SMIN=0.05 RPSQ=15}
DIELECTRIC D0 { THICKNESS = 0.141 ER=4 }
```


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Introduction to Chip Finish : STAR RC

- **Layer map file**
  - **What is Layer map file?**
    - Map file maps the .ITF file & .tf file of the layer and via names
    - The map file matches the layer and via names in the Milkyway technology file with the names in the ITF (Interconnect Technology Format) file
  - **Structure of Layer map file**

```
conducting_layers
PO                  POLY
M1                  M1
M2                  M2
M3                  M3
M4                  M4
M5                  M5
M6                  M6
M7                  M7
M8                  M8
M9                  M9
MRDL                MRDL
```

Layer names

```
via_layers
VIA1                VIA1
VIA2                VIA2
VIA3                VIA3
VIA4                VIA4
VIA5                VIA5
VIA6                VIA6
VIA7                VIA7
VIA8                VIA8
VIARDL              VIARDL

remove_layers
CO                  POLYCONT
```

VIA names

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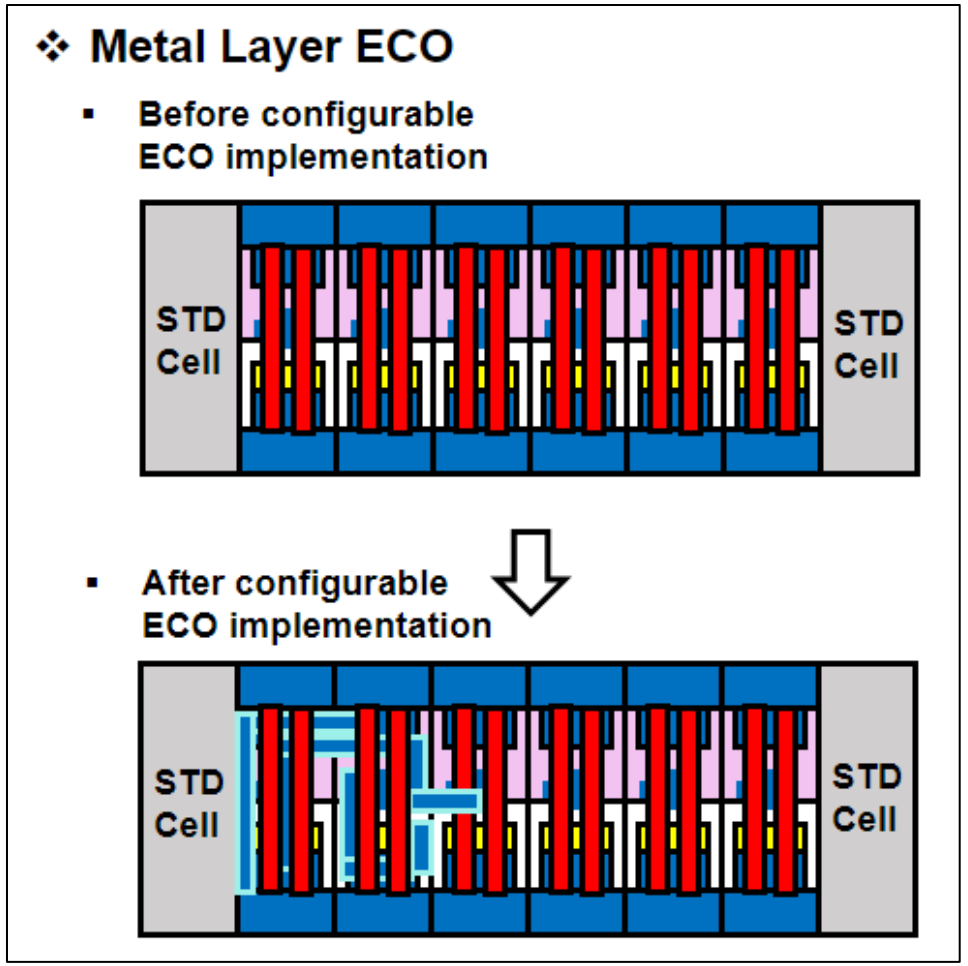
Introduction to Chip Finish : ECO

- **ECO**
  - **What is ECO (Engineering Change Orgder) ?**
    - ECO is technique to add or remove the logic with minimum modifications in the design
    - For fixing post Synthesis, Route, Silicon issues during P&R flow
    - Fixing both timing and functionally issues
    - Spare Cells placed in the design are used for ECO
    - A Logic Gate/Flip-Flop can be realized using these spare Cells
    - Different flavors of Gate, of required drive strength can be realized any where in the design
    - Only Metal/Contact changes are need after fixing the defective design

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Introduction to Chip Finish : ECO

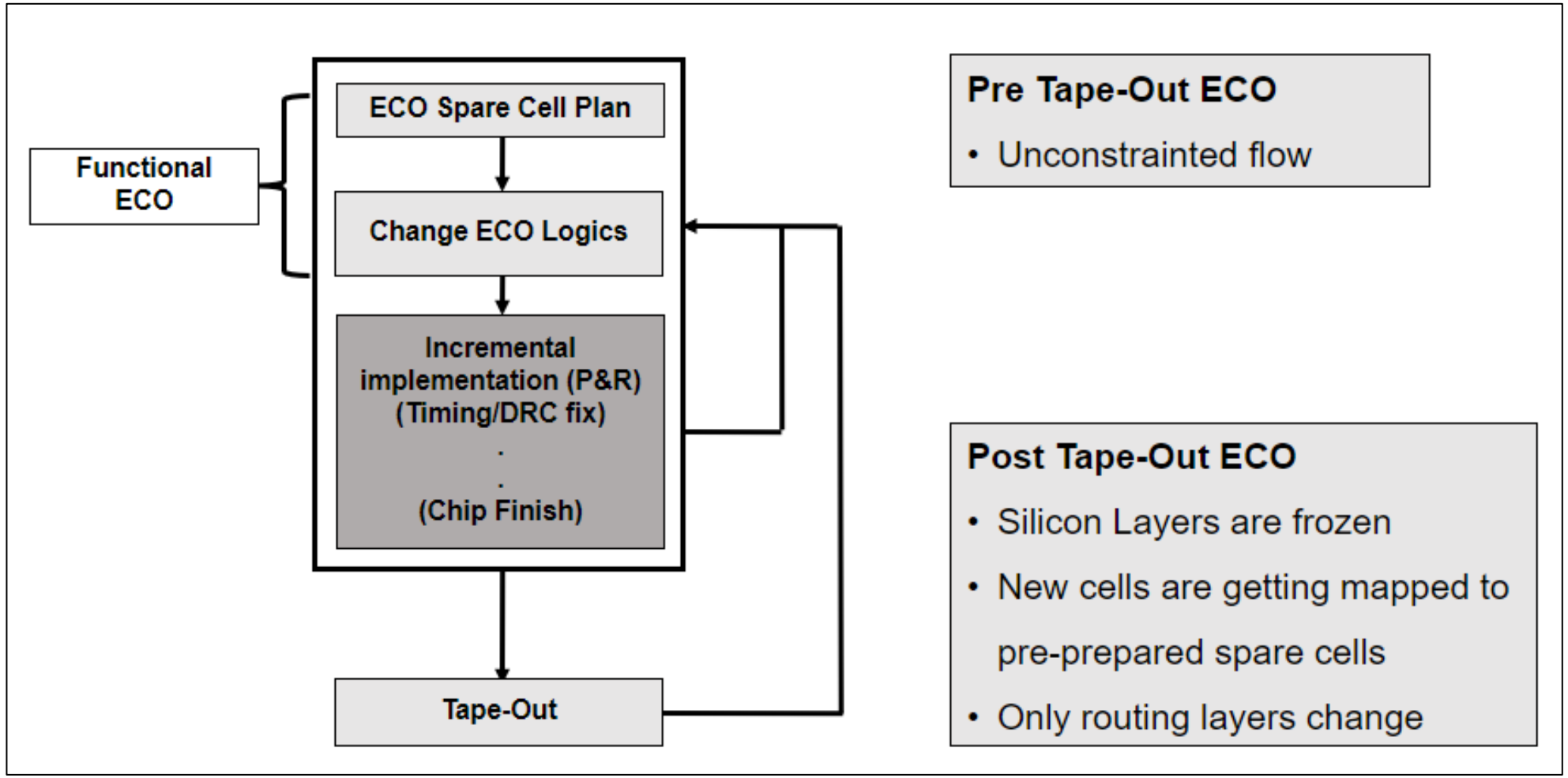
- **Types of ECO (Engineering Change Order)**
  - **Post Synthesis ECO (after synthesis)**
    - ECO with Synthesized Netlist
  - **Post Route ECO (after P&R flow)**
    - During minute change in the design after full Tape-out is over
    - Use spare cells & Metal layers only
    - Metal Layer ECO
      - Only Metal Layer changes
      - During minute change in RTL after Active/Base Layer Tape-out is over
      - Cleaning-up routing for Signal Integrity (SI)
    - Active Layer ECO
      - Use Spare cells
      - During minute change in RTL just after Routing is over
      - NAND Gate based spare cell can be used to realize the new ECO logic
  - **Post Synthesis ECO (after synthesis)**
    - Use spare cells & Metal Layers
    - To recover from minute manufacturing issues



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Introduction to Chip Finish : ECO

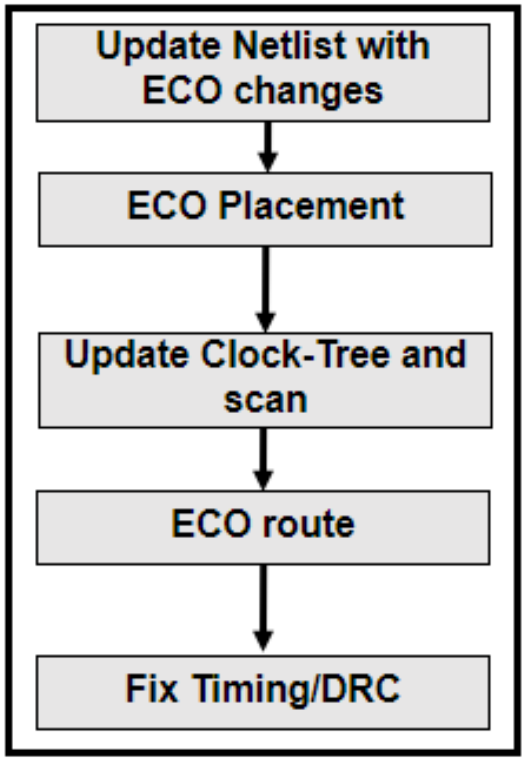
- **Pre & Post Tape-out ECO**



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Introduction to Chip Finish : ECO

- **Flow of unconstrained Pre Tape-Out ECO**

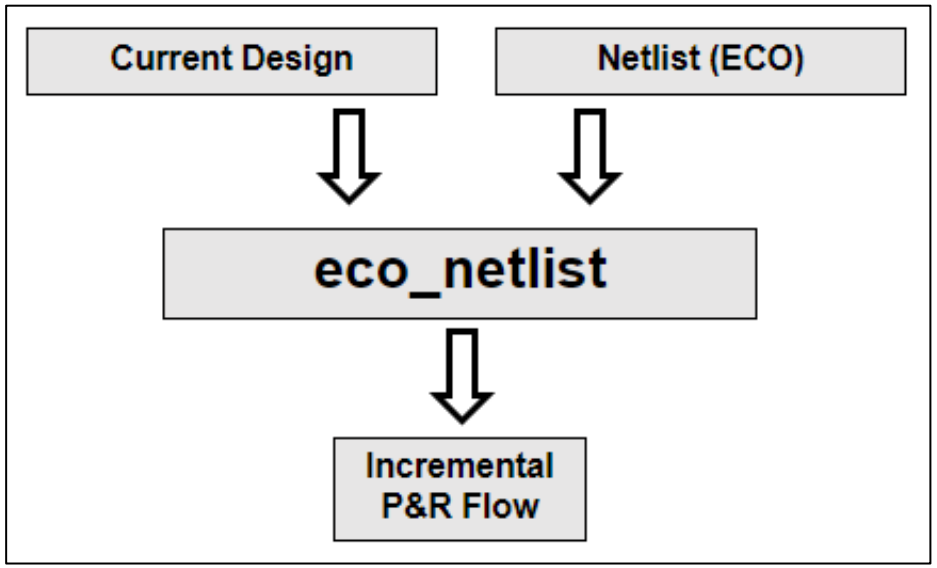


1. **Update Netlist with ECO changes**
   - Tcl commands or scripts
   - "eco_netlist" for verilog with ECO changes
2. **Place ECO cells**
   - Place_eco_cells
3. **Update clock and scan (if Flip-Flops are added or removed)**
   - Manual steps (use "connect_net" or "disconnect_net")
4. **Route new ECO nets**
   - Route_eco
5. **Fix Timing/DRC violation**
   - Route_opt
   - Search & repair

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Introduction to Chip Finish : ECO

- **Flow of unconstrained Pre Tape-Out ECO**
  - **"eco_netlist" identifies changes between two netlist versions**
    - A netlist editing script is generated, which can then be applied to the design
    - Applied at the module level
  - **Implementation of newly inserted ECO cells occurs through incremental P&R flow**



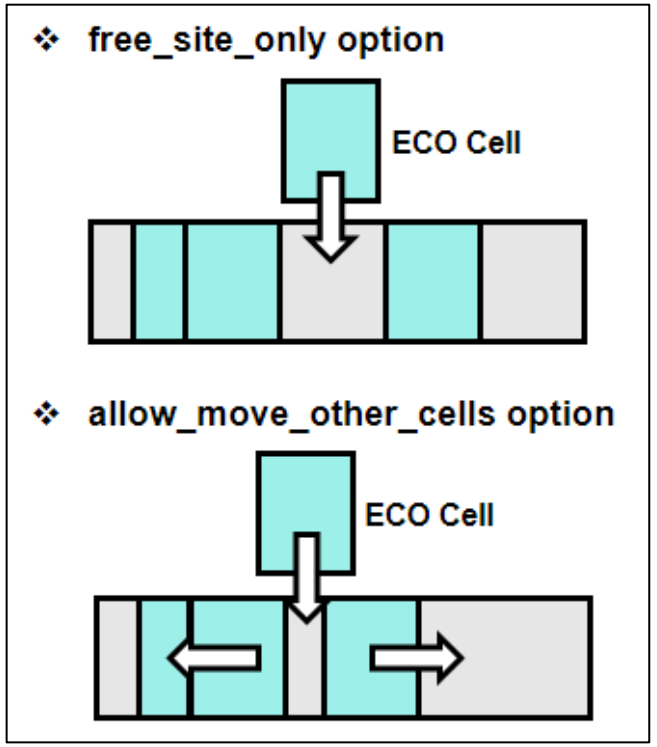
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Introduction to Chip Finish : ECO

- **Flow of unconstrained Pre Tape-Out ECO**
  - **ECO placement**

```
place_eco_cells -eco_changed_cells \
                -legalize_mode free_site_only |
                               allow_move_other_cells |
                               minimun_physical_impact
```

- **free_site_only (default)**
  - ECO cells are legalized on free sites without moving preexisting cells
- **allow_move_other_cells**
  - ECO cells are legalized to their nearest legal location by moving preexisting cells
- **minimum_physical_impact**
  - Combines both ECO cell legalizations into a single-pass legalization
  - Valid only with "-legalize_only"
  - To be used with the "Physically-aware ECO flow with PrimeTime"



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Introduction to Chip Finish : ECO

- **ECO Route**
  - **ECO route is automatically performed after optimization is done**
    - By default, 5 iterations of route detail are performed
    - You can increase the number of iterations of routing inside "route_opt" command

```
set_app_options -name route.detail.eco_max_number_of_iterations \
                -value 50
```

  - **Or, apply additional app_optins, route_detail after route_opt command**

```
reoute_detail -incremental -max_number_iterations 50
```

  - **The "route_eco" command performs ECO routing**

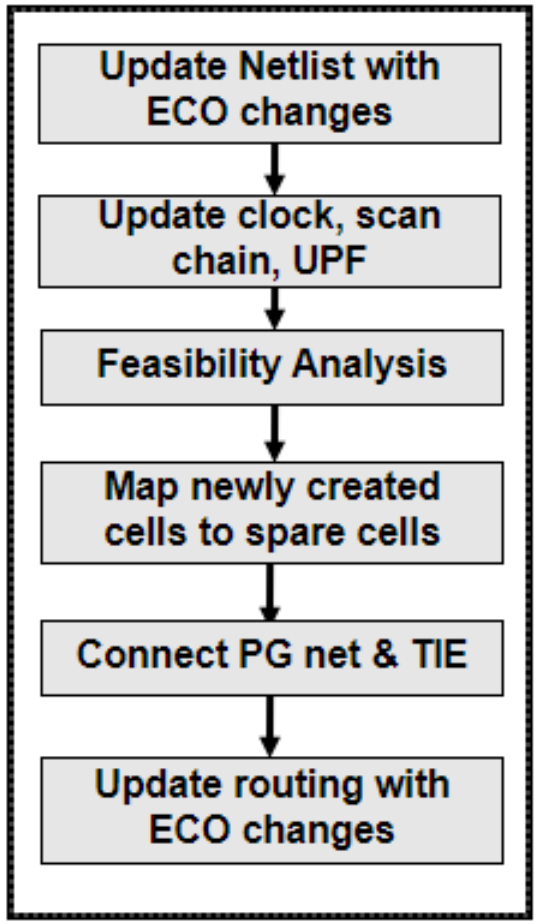
```
route_eco -max_detail_route_iterations 50
```

    - It first connects the open nets and then fixes the DRC violations
    - The options control which nets are connected and in which areas of the chip the DRC violations are fixed
    - Use this command only after signal routing has been completed
    - To route nets before signal routing, use the route_group command

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Introduction to Chip Finish : ECO

- **Flow of Freeze Silicon ECO Flow**

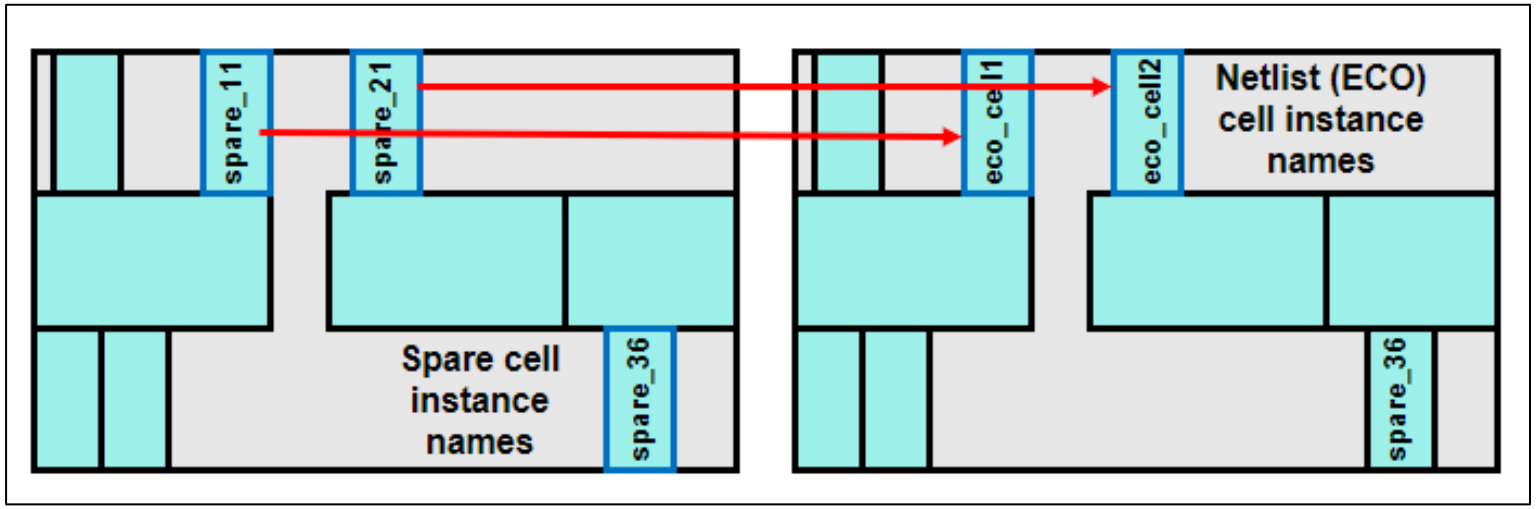


1. **Insert & place spare cells (after placement**
2. **Update netlist with ECO changes in freeze-silicon mode**
   - set_app_options –list {design.eco_freeze_silicon_mode true}
   - With tcl editing commands
   - eco_netlist + source
3. **Manually update clock tree, scan chain, UPF file**
4. **Feasibility Analysis**
   - check_freeze_silicon
5. **Map newly created cells to spare cells**
   - place_freeze_silicon (automatic)
   - map_freeze_silicon (manual)
6. **Connect PG, Connect tie-off pins to the tie cells**
   - connect_pg_net
   - connect_freeze_silicon_tie_cells
7. **Connect PG, Connect tie-off pins to the tie cells**
   - route_eco
   - Search & repair

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Introduction to Chip Finish : ECO

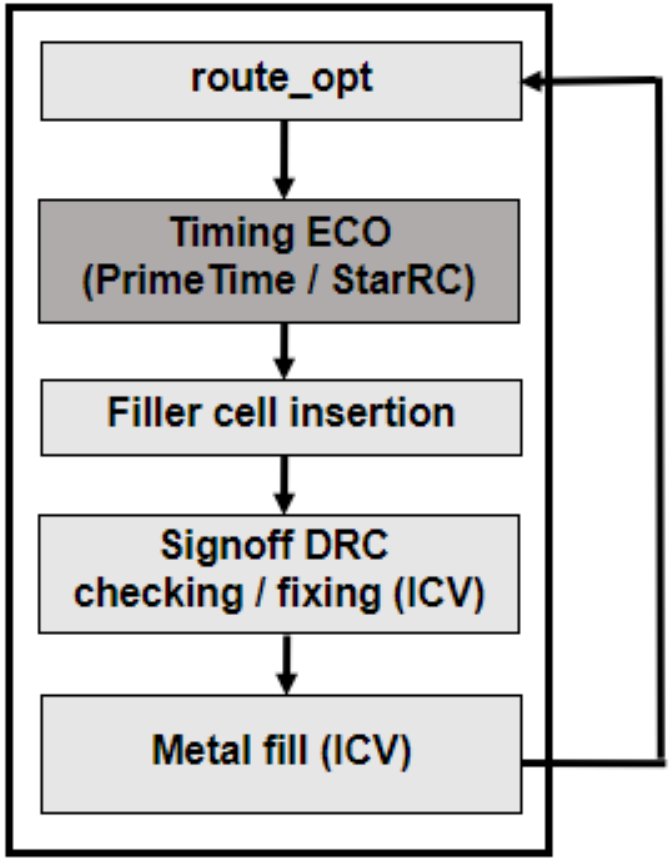
- **Map ECO Cells to Spare Cells**
  - **Placement based mapping (place_freeze_silicon)**
    - Coarse placement on ECO cells
    - Map ECO cells to the closest compatible spare cells (same reference name)
  - **Manual mapping (map_freeze_silicon)**
  - **Resized or removed cells are recycled as spare cells**



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Introduction to Chip Finish : ECO

- **Timing ECO**



- **ECO file**
  - Eco file contains a series of changes required in the form of P&R tool (ICC2) command for fixing the issue
  - Based on the analysis, sometimes generate the eco file from the signoff tool itself or sometimes create own eco file
  - If we fix the setup timing, we need to upsize cells (cell sizing) or convert lower vt cells (vth swap)
  - Signoff tool will generate commands for each cell that need to be changed and write in a file
  - Source ECO file in P&R tool (ICC2) followed by refine placement and "eco_route"
- **ECO implementation**
  - ECO implementation is generally done in the batch mode of the tool
  - Delete the Filler cells before souring the eco file
  - Once the eco file is sourced, all the required changes is done
  - After sourcing eco file, there might be a change in the size of cells or addition/delection of some cells so we need to do refine placement to followed by the eco_route
  - The updated db need to be saved for next stage of ECO or the final db in case of all issues are fixed

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WEEK 15 : LAB for Chip Finish

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Chip Finish : Introduction to placement.tcl

- Review and check the variables in each scripts
  - **Go to the Script Directory**

```
  < user@linux script >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/07_chip_finish/script
  < user@linux script >> ls
01_design_stup_for_chip_finish.tcl  03_write_output_file.tcl     full_script
02_insert_std_filler.tcl            04_extract_spef_in_ICC2.tcl
```

  - **Review 01_design_setup_for_chip_finish.tcl**

```
#made by kjinsoo
# module load
#module load icc2/2021.06-SP5
# open icc2_shell
#icc2_shell -gui | tee -i ./09_chip_finish.log

#set current step & before step
set step chip_finish
set before_step route_opt_fix
#source design_setup.tcl
source -e ./icc2_common_env/design_setup.tcl

#source timer.rpt script - report runtime for placement
source ./icc2_common_env/sec_runtime.tcl

sec_StartTimer copy_lib

#set_host_options -max_cores 16

#open lib
open_lib ../06_route/nlib/ORCA_${before_step}.nlib
#copy lib
copy_lib -from ../06_route/nlib/ORCA_${before_step}.nlib -to ./nlib/ORCA_${step}.nlib
#close & open current lib
close_lib -all
open_lib ./nlib/ORCA_${step}.nlib
#open block
open_block ./nlib/ORCA_${step}.nlib:ORCA.design
#link_block
link_block -rebind
```


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Chip Finish : Introduction to placement.tcl

- Review and check the variables in each scripts
  - **Review 02_insert_std_filler.tcl**

```
####################
#insert_filler
####################

set SH_FILLERS "*/SHFILL128_HVT */SHFILL64_HVT */SHFILL3_HVT */SHFILL2_HVT */SHFILL1_HVT"
create_stdcell_fillers -lib_cells $SH_FILLERS -rules {post_route_auto_delete} ;# it allows route_opt & eco_opt to automatically remove filler cells

#connect pg_net for all STD filler cell (VDD, VSS)
set FILLERS [get_cells -filter "full_name =~ *SHFILL*"]

set i 0

foreach x $FILLERS {
set pins [get_pins -of_objects $x -filter "port_type == power"]
connect_pg_net -net VDD [get_pins $pins]

incr i
}

foreach x $FILLERS {
set pins [get_pins -of_objects $x -filter "port_type == ground"]
connect_pg_net -net VSS [get_pins $pins]

incr i
}

#remove Filler Cell with violation
remove_stdcell_fillers_with_violation
```


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Chip Finish : Introduction to placement.tcl

- Review and check the variables in each scripts
  - **Review 03_write_output_file.tcl**

```
######################
#write Netlist file
######################

write_verilog ./output/ORCA.v -exclude {leaf_module_declarations corner_cells pad_spacer_cells filler_cell
s flip_chip_pad_cells empty_modules unconnected_ports spare_cells}

######################
#write gds
######################

write_gds ./output/ORCA.gds -exclude_empty_block -skip_unplaced_cells -lib_cell_view layout


######################
#write def
######################

write_def ./output/ORCA.def -include_tech_via_definitions -include {rows_tracks vias cells ports blockages
 bounds specialnets nets routing_rules}
```

  - **Review 04_extract_spef_in_ICC2.tcl**

```
######################
#extract_spef (ICC2)
######################
write_parasitics -output ./output/ORCA

#save & exit

save_lib -all

exit
```


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Chip Finish: Use ICC2 (IC Compiler II tool)

It is also good to Read Design by source 02_floorplan.tcl, but when you proceed for the first time, it is recommended that you open the script and apply it one phrase at a time

1. Invoke tool icc2_shell

- **Recommend the icc2/2021.06-SP5 version**

>> icc2_shell –gui | tee -i ./09_chip_finish.log ;# (| tee -I ./09_chip_finish.log) is the command to output log

```
 < user@linux 07_chip_finish >> icc2_shell -gui | tee -i ./09_chip_finish.log



                              IC Compiler II (TM)

            Version S-2021.06-SP5 for linux64 - Jan 14, 2022 -SLE

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Loading user preference file /u/kjinsoo/.synopsys_icc2_gui/preferences.tcl
Load ICV ICCII menu file: /global/apps/icv_2022.12-SP5/etc/tcl-u/Icc2Menu.tcl
 + VUE INFO: Please click View->IC Validator VUE in LayoutWindow menu
             to launch VUE.

 + VUE INFO: Found a usable port: 2446

icc2_shell>
```

2. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/07_chip_finish/script/01_design_setup_for_chip_finish.tcl

- **set step & source design_setup.tcl**

```
icc2_shell> set step chip_finish
chip_finish
icc2_shell> set before_step route_opt_fix
route_opt_fix
icc2_shell> source -e ./icc2_common_env/design_setup.tcl
```


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Chip Finish: Use ICC2 (IC Compiler II tool)

2. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/07_chip_finish/script/01_design_setup_for_chip_finish.tcl

- **open lib**

```
icc2_shell> open_lib ../06_route/nlib/ORCA_${before_step}.nlib
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir
/06_route/nlib/ORCA_route_opt_fix.nlib' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir
/00_create_ndm/ndm/EXPLORE_physical_only.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir
/00_create_ndm/ndm/saed32hvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir
/00_create_ndm/ndm/saed32io_fc.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir
/00_create_ndm/ndm/saed32lvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir
/00_create_ndm/ndm/saed32pll.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir
/00_create_ndm/ndm/saed32rvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir
/00_create_ndm/ndm/saed32sram.ndm' (FILE-007)
{ORCA_route_opt_fix.nlib}
```

Check that ndm is properly apllied each time you open the nlib

- **Copy ORCA_route_opt_fix.nlib to ORCA_chip_finish.nlib & open ORCA_chip_finish.nlib**

```
copy_lib -from ../06_route/nlib/ORCA_${before_step}.nlib -to ./nlib/ORCA_${step}.nlib
Information: Copying from library "ORCA_route_opt_fix.nlib" to "./nlib/ORCA_chip_finish.nlib".
{ORCA_chip_finish.nlib}
icc2_shell> close_lib -all
Closing all libraries...
1
icc2_shell> open_lib ./nlib/ORCA_${step}.nlib
```


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Chip Finish: Use ICC2 (IC Compiler II tool)

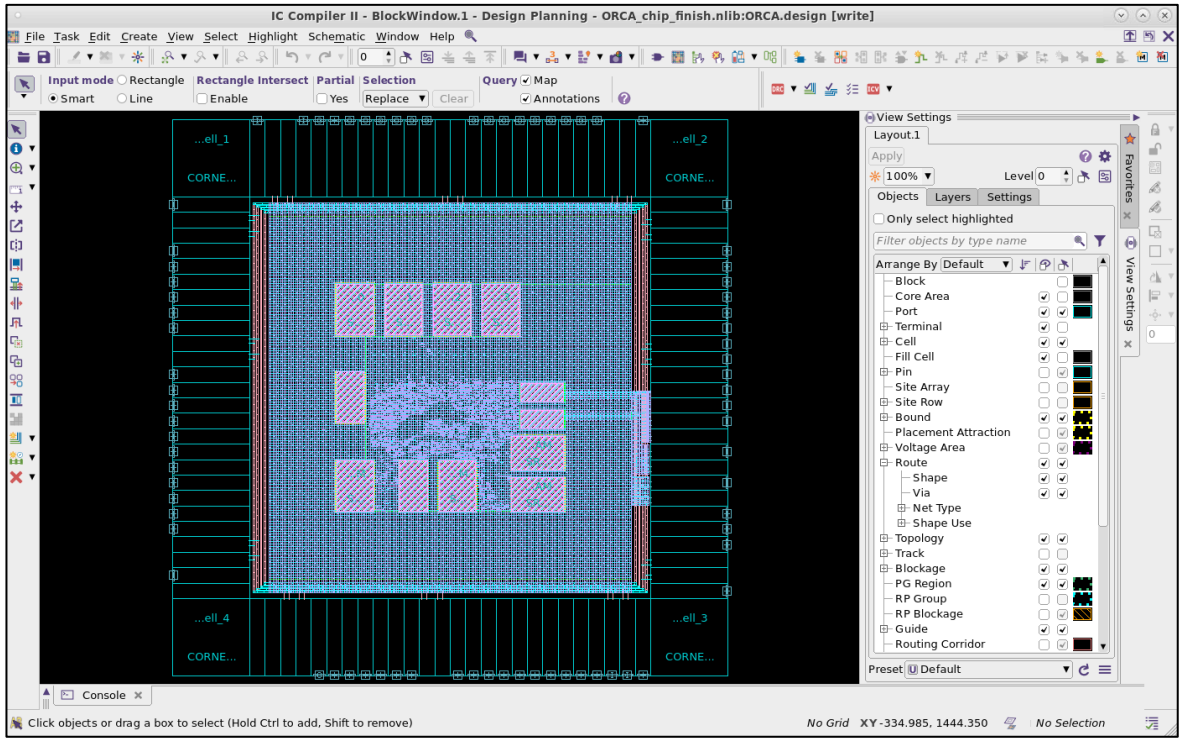
2. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/07_chip_finish/script/01_design_setup_for_chip_finish.tcl

- **open block**

```
icc2_shell> open_block ./nlib/ORCA_${step}.nlib:ORCA.design
Information: User units loaded from library 'saed32hvt' (LNK-040)
Opening block 'ORCA_chip_finish.nlib:ORCA.design' in edit mode
{ORCA_chip_finish.nlib:ORCA.design}
icc2_shell> link_block
Using libraries: ORCA_chip_finish.nlib EXPLORE_physical_only saed32hvt saed32io_fc saed32lvt s
aed32pll saed32rvt saed32sram
```

Check using libraries

- **Check the copied block in the gui window**



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Chip Finish: Use ICC2 (IC Compiler II tool)

3. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/07_chip_finish/script/02_insert_std_filler.tcl

- **Insert STD Filler cells in Core Area**

```
icc2_shell> set SH_FILLERS "*/SHFILL128_HVT */SHFILL64_HVT */SHFILL3_HVT */SHFILL2_HVT */SHFI
LL1_HVT"
*/SHFILL128_HVT */SHFILL64_HVT */SHFILL3_HVT */SHFILL2_HVT */SHFILL1_HVT
icc2_shell> create_stdcell_fillers -lib_cells $SH_FILLERS -rules {post_route_auto_delete}
INFO:: begin filler insertions...
CHF::NPL_CHKQUERY_PNET checking turned off by default
master SHFILL128_HVT has site unit
master SHFILL64_HVT has site unit
master SHFILL3_HVT has site unit
master SHFILL2_HVT has site unit
master SHFILL1_HVT has site unit
 Use site unit (152)
Warning: Routing direction of metal layer PO is neither "horizontal" nor "vertical".  PDC che
cks will not be performed on this layer. (PDC-003)
PDC app_options settings =========
        place.legalize.enable_prerouted_net_check: 1
        place.legalize.num_tracks_for_access_check: 1
        place.legalize.use_eol_spacing_for_access_check: 0
        place.legalize.allow_touch_track_for_access_check: 1
        place.legalize.reduce_conservatism_in_eol_check: 0
        place.legalize.preroute_shape_merge_distance: 0.0
        place.legalize.enable_non_preferred_direction_span_check: 0

Layer M1: cached 0 shapes out of 8777 total shapes.
Layer M2: cached 664 shapes out of 73875 total shapes.
Cached 86128 vias out of 274840 total vias.
CHF: loading design...
CHF:: Loading blkInsts from NDM...
CHF:: Building row/inst relations...
INFO:: Use fillers in order(1000): SHFILL128_HVT (128 x 1), SHFILL64_HVT (64 x 1), SHFILL3_HV
T (3 x 1), SHFILL2_HVT (2 x 1), SHFILL1_HVT (1 x 1),
        20% complete ...
        30% complete ...
        40% complete ...
        50% complete ...
        60% complete ...
        70% complete ...
        80% complete ...
Regular Filler Insertion Complete
CHF:: Create all new fillers...
... 18598 of regular filler SHFILL128_HVT inserted
... 1332 of regular filler SHFILL64_HVT inserted
... 63315 of regular filler SHFILL3_HVT inserted
... 3829 of regular filler SHFILL2_HVT inserted
... 3892 of regular filler SHFILL1_HVT inserted
```

Check Filler cells after inserted

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Chip Finish: Use ICC2 (IC Compiler II tool)

3. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/07_chip_finish/script/02_insert_std_filler.tcl

- **Connect_pg_net for all STD FIller Cells (VDD, VSS)**

```
icc2_shell> set FILLERS [get_cells -filter "full_name =~ *SHFILL*"]
```

```
icc2_shell> set i 0
0
icc2_shell>
icc2_shell> foreach x $FILLERS {
? set pins [get_pins -of_objects $x -filter "port_type == power"]
? connect_pg_net -net VDD [get_pins $pins]
?
? incr i
? }
icc2_shell>
icc2_shell> foreach x $FILLERS {
? set pins [get_pins -of_objects $x -filter "port_type == ground"]
? connect_pg_net -net VSS [get_pins $pins]
?
? incr i
? }
```

- **Remove STD Filler Cells with violation**

```
icc2_shell> remove_stdcell_fillers_with_violation
Information: The command 'remove_stdcell_fillers_with_violation' cleared the undo history. (U
NDO-016)
Cell Min-Routing-Layer = M1
Cell Max-Routing-Layer = M6
```

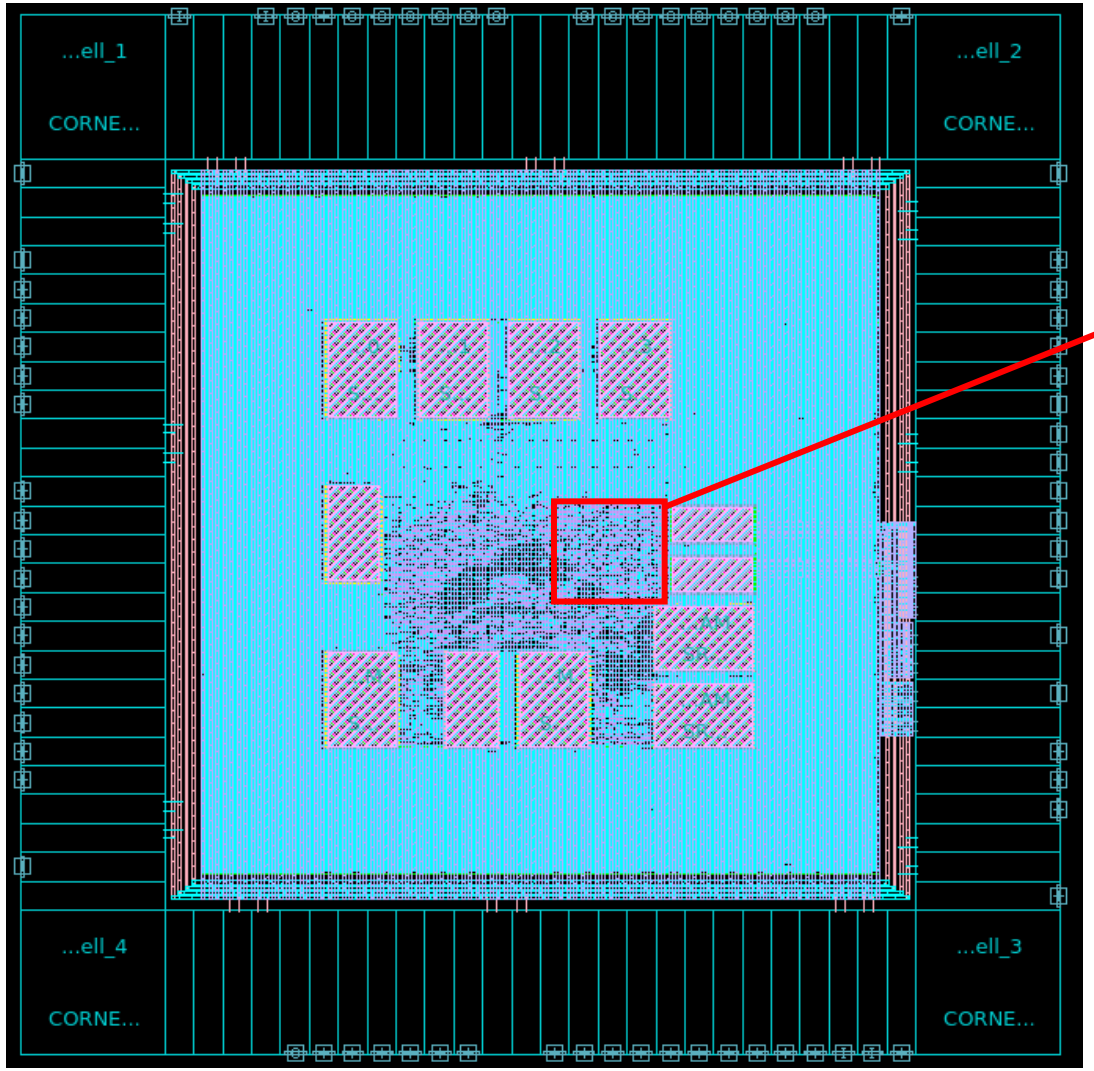
```
Partition 392, Fillers with Violations = 0
[End Removing Filler Cells] Elapsed real time: 0:00:14
[End Removing Filler Cells] Elapsed cpu  time: sys=0:00:00 usr=0:00:13 total=0:00:13
[End Removing Filler Cells] Stage (MB): Used   40  Alloctr   41  Proc  280
[End Removing Filler Cells] Total (MB): Used  150  Alloctr  151  Proc 5190
Updating the database ...
Deleted 0 cell instances
```

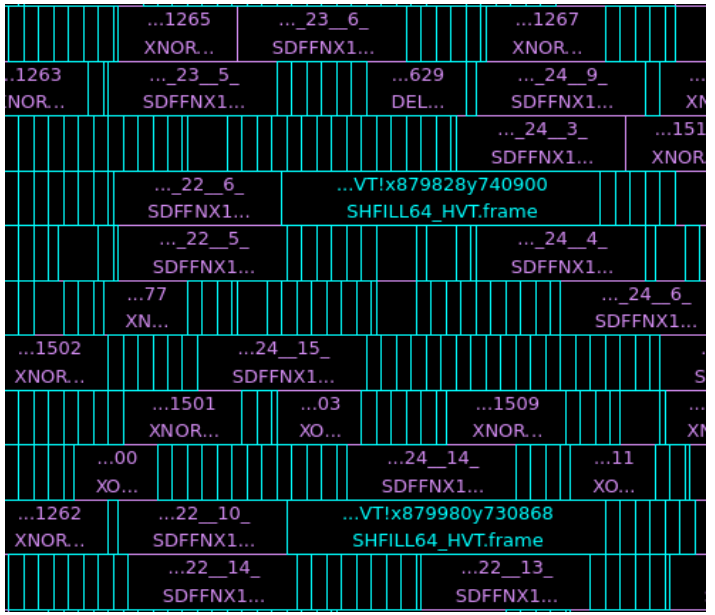

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Chip Finish: Use ICC2 (IC Compiler II tool)

3. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/07_chip_finish/script/02_insert_std_filler.tcl

- **Check STD Filler Cells after inserted**





There should be no space between STD Cell and STD FILER Cell

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Chip Finish: Use ICC2 (IC Compiler II tool)

4. **Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/07_chip_finish/script/03_write_output_file.tcl**
   - **Write output file after insert STD filler cells (Netlist, GDS, DEF)**

```
icc2_shell> write_verilog ./output/ORCA.v -exclude {leaf_module_declarations corner_cells pad_spacer_cells
 filler_cells flip_chip_pad_cells empty_modules unconnected_ports spare_cells}
1
icc2_shell> write_gds ./output/ORCA.gds -exclude_empty_block -skip_unplaced_cells -lib_cell_view layout
```

```
icc2_shell> write_def ./output/ORCA.def -include_tech_via_definitions -include {rows_tracks vias cells por
ts blockages bounds specialnets nets routing_rules}
****************************************
Report : Data Mismatches
Version: S-2021.06-SP5
Date   : Tue Dec 12 11:54:57 2023
****************************************
No mismatches exist on the design.
--------------------------------------
Number of Written DEF Constructs
--------------------------------------
VERSION                      : 1
DIVIDERCHAR                  : 1
BUSBITCHARS                  : 1
DESIGN                       : 1
UNITS                        : 1
DIEAREA                      : 1
ROW                          : 561
TRACKS                       : 20
VIAS                         : 78
NONDEFAULTRULES              : 2
COMPONENTS                   : 101969
PINS                         : 78
BLOCKAGES                    : 119
SPECIALNETS                  : 7697
NETS                         : 13984
1
```

   - **Check output files in $WORK_DIR/2_BACKEND/1_pnr/rundir/07_chip_finish/output/***

```
  < user@linux output >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/07_chip_finish/output
  < user@linux output >> ls
ORCA.def  ORCA.gds  ORCA.v
```


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Chip Finish: Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/07_chip_finish/script/04_extract_spef_in_ICC2.tcl

- **Extract .spef files**

```
icc2_shell> write_parasitics -output ./output/ORCA
Information: The command 'write_parasitics' cleared the undo history. (UNDO-016)
Information: Design ORCA has 13690 nets, 0 global routed, 13612 detail routed. (NEX-024)
NEX: extract design ORCA
Information: The RC mode used is DR for design 'ORCA'. (NEX-022)
---extraction options---
Corner: ff_n40c
 late_cap_scale                : 1
 late_res_scale                : 1
 early_cap_scale               : 1
 early_res_scale               : 1
Corner: ss_125c
 late_cap_scale                : 1
 late_res_scale                : 1
 early_cap_scale               : 1
 early_res_scale               : 1
Corner: ff_125c
 late_cap_scale                : 1
 late_res_scale                : 1
 early_cap_scale               : 1
 early_res_scale               : 1
Corner: ss_n40c
 late_cap_scale                : 1
 late_res_scale                : 1
 early_cap_scale               : 1
 early_res_scale               : 1
```

```
Global options:
 reference_direction       : use_from_tluplus
 real_metalfill_extraction : none
 virtual_shield_extraction : true
---app options---
 host.max_cores                  : 1
 extract.connect_open            : true
 extract.incremental_extraction : true
 extract.enable_coupling_cap    : false
Extracting design: ORCA
Information: coupling capacitance is lumped to ground. (NEX-030)
Information: 13612 nets are successfully extracted. (NEX-028)
Information: batch extract takes 9.28 seconds, elapse 9.84 seconds (NEX-015)
Information: The stitching and editing of coupling caps is turned OFF for design 'ORCA_chip_finish.nlib:OR
CA.design'. (TIM-125)
NEX: write corner ID 0 parasitics to ./output/ORCA.minTLU_-40.spef
spefName ./output/ORCA.minTLU_-40.spef, corner name ff_n40c
NEX: write corner ID 1 parasitics to ./output/ORCA.maxTLU_125.spef
spefName ./output/ORCA.maxTLU_125.spef, corner name ss_125c
NEX: write corner ID 2 parasitics to ./output/ORCA.minTLU_125.spef
spefName ./output/ORCA.minTLU_125.spef, corner name ff_125c
NEX: write corner ID 3 parasitics to ./output/ORCA.maxTLU_-40.spef
spefName ./output/ORCA.maxTLU_-40.spef, corner name ss_n40c
NEX:Warning write_parasitics supports only Ohm and kOhm as user resistance unit not 1MOhm, we force to set
 it Ohm.Information: Design Average RC for design ORCA  (NEX-011)
Information: r = 1.787804 ohm/um, via_r = 0.502450 ohm/cut, c = 0.069288 ff/um, cc = 0.000000 ff/um (X dir
) (NEX-017)
Information: r = 1.785714 ohm/um, via_r = 0.629701 ohm/cut, c = 0.081358 ff/um, cc = 0.000000 ff/um (Y dir
) (NEX-017)
NEX: write_parasitics command finished
```

- **Save & Exit**

```
icc2_shell> save_lib -all
Saving all libraries...
1
icc2_shell> exit
```


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Chip Finish: Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/07_chip_finish/script/04_extract_spef_in_ICC2.tcl

- **Check .spef files in $WORK_DIR/2_BACKEND/1_pnr/rundir/07_chip_finish/output/***

```
  < user@linux output >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/07_chip_finish/output
  < user@linux output >> ls
ORCA.def  ORCA.maxTLU_125.spef  ORCA.minTLU_125.spef  ORCA.spef_scenario
ORCA.gds  ORCA.maxTLU_-40.spef  ORCA.minTLU_-40.spef  ORCA.v
```

- **Open *"ORCA_TOP.spef_scenario"* and check scenarios for all .spef files**

```
./output/ORCA.minTLU_-40.spef for scenarios:
        scenario: func.ff_n40c, corner type: late/early
        scenario: scan_capture.ff_n40c, corner type: late/early
        scenario: scan_shift.ff_n40c, corner type: late/early

./output/ORCA.maxTLU_125.spef for scenarios:
        scenario: func.ss_125c, corner type: late/early
        scenario: scan_capture.ss_125c, corner type: late/early
        scenario: scan_shift.ss_125c, corner type: late/early

./output/ORCA.minTLU_125.spef for scenarios:
        scenario: func.ff_125c, corner type: late/early
        scenario: scan_capture.ff_125c, corner type: late/early
        scenario: scan_shift.ff_125c, corner type: late/early

./output/ORCA.maxTLU_-40.spef for scenarios:
        scenario: func.ss_n40c, corner type: late/early
        scenario: scan_capture.ss_n40c, corner type: late/early
        scenario: scan_shift.ss_n40c, corner type: late/early
```


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WEEK 15 : LAB for STAR RC

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STAR RC : Introduction to star.cmd & :run

- Review and check the variables in each scripts
  - **Go to the Script Directory**

```
  < user@linux STAR_RC >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/STAR_RC
  < user@linux STAR_RC >> ls
CAPnNETLIST  MAP  MAX_bst  MAX_wst  MIN_bst  MIN_wst  NXTGRD
```

  - **Review $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MAX_bst/star.cmd**

```
************
* Database *
************
NDM_DATABASE : ../../rundir/07_chip_finish/nlib/ORCA_chip_finish.nlib
BLOCK : ORCA
* MILKYWAY_DATABASE :
* LEF_FILE: %LEF%
* TOP_DEF_FILE: %Def%
* LEF_USE_OBS: YES
* METAL_FILL_GDS_FILE :                %DMGDS:i,# Dummy metal GDS file%
* GDS_LAYER_MAP_FILE :                 %GdsMap%
* METAL_FILL_POLYGON_HANDLING:         FLOATING

**************
* Extraction *
**************
  MAPPING_FILE : ../MAP/ORCA.map
  TCAD_GRD_FILE : ../NXTGRD/saed32nm_1p9m_Cmax.nxtgrd
*  MODE: 400
  COUPLE_TO_GROUND : NO
  EXTRACT_VIA_CAPS: YES
  INTRANET_CAPS:    NO
  REDUCTION: YES
  REDUCTION_MAX_DELAY_ERROR:           5.0e-13
*  SENSITIVITY: YES
  TEMPERATURE_SENSITIVITY: YES
  OPERATING_TEMPERATURE : -40
```

Extract the .spef files based on the NDM Database & Block Name

Extraction using layer.map file & nxtgrd file

Operation Temperature is -40 in MAX_bst

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STAR RC : Introduction to star.cmd & :run

- Review and check the variables in each scripts
  - **Review $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MAX_bst/star.cmd**

```
**************
* Processing *
**************
*  NUM_PARTS: 2
* REMOVE_FLOATING_NETS: YES
* REMOVE_DANGLING_NETS: YES
* SKIP_CELLS: !
* SKIP_CELLS_FILE:

**************
* Netlisting *
**************
* NETLIST_NAME_MAP: NO
  NETLIST_FILE : ../CAPnNETLIST/ORCA_MAX_n40c.spef.gz
  NETLIST_FORMAT : SPEF
*  NETLIST_POSTPROCESS_COMMAND: gzip -f

*********
* Noise *
*********
* COUPLING_REPORT_FILE:               coupling.rpt
* COUPLING_REPORT_NUMBER:             1000000
*  COUPLING_ABS_THRESHOLD:             1e-15
*  COUPLING_REL_THRESHOLD:             0.02
```

Path from which the .spef file is extracted & .spef.gz file name

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STAR RC : Introduction to star.cmd & :run

- Review and check the variables in each scripts
  - **Review $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MAX_bst/:run**

```
#module load starrc/2021.06-SP5-5
#StarXtract -clean star.cmd&
#sleep 60
StarXtract star.cmd
```

Run STAR RC based on star.cmd file

  - **Review $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MAX_wst/star.cmd**

```
************
* Database *
************
NDM_DATABASE : ../../rundir/07_chip_finish/nlib/ORCA_chip_finish.nlib
BLOCK : ORCA
* MILKYWAY_DATABASE :
* LEF_FILE: %LEF%
* TOP_DEF_FILE: %Def%
* LEF_USE_OBS: YES
* METAL_FILL_GDS_FILE :                %DMGDS:i,# Dummy metal GDS file%
* GDS_LAYER_MAP_FILE :                 %GdsMap%
* METAL_FILL_POLYGON_HANDLING:         FLOATING

**************
* Extraction *
**************
  MAPPING_FILE : ../MAP/ORCA.map
  TCAD_GRD_FILE : ../NXTGRD/saed32nm_1p9m_Cmax.nxtgrd
*   MODE: 400
  COUPLE_TO_GROUND : NO
  EXTRACT_VIA_CAPS: YES
  INTRANET_CAPS:    NO
  REDUCTION: YES
  REDUCTION_MAX_DELAY_ERROR:           5.0e-13
*   SENSITIVITY: YES
  TEMPERATURE_SENSITIVITY: YES
  OPERATING_TEMPERATURE : 125
```

Extract the .spef files based on the NDM Database & Block Name

Extraction using layer.map file & nxtgrd file

Operation Temperature is 125 in MAX_wst

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STAR RC : Introduction to star.cmd & :run

- Review and check the variables in each scripts
  - **Review $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MAX_wst/star.cmd**

```
**************
* Processing *
**************
*  NUM_PARTS: 2
* REMOVE_FLOATING_NETS: YES
* REMOVE_DANGLING_NETS: YES
* SKIP_CELLS: !
* SKIP_CELLS_FILE:

**************
* Netlisting *
**************
* NETLIST_NAME_MAP: NO
  NETLIST_FILE : ../CAPnNETLIST/ORCA_MAX_125c.spef.gz
  NETLIST_FORMAT : SPEF
*  NETLIST_POSTPROCESS_COMMAND: gzip -f

*********
* Noise *
*********
* COUPLING_REPORT_FILE:               coupling.rpt
* COUPLING_REPORT_NUMBER:             1000000
*  COUPLING_ABS_THRESHOLD:             1e-15
*  COUPLING_REL_THRESHOLD:             0.02
```

Path from which the .spef file is extracted & .spef.gz file name

  - **Review $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MAX_wst/:run**

```
#module load starrc/2021.06-SP5-5
#StarXtract -clean star.cmd&
#sleep 60
StarXtract star.cmd
```

Run STAR RC based on star.cmd file

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STAR RC : Introduction to star.cmd & :run

- Review and check the variables in each scripts
  - **Review $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MIN_bst/star.cmd**

```
************
* Database *
************
NDM_DATABASE : ../../rundir/07_chip_finish/nlib/ORCA_chip_finish.nlib
BLOCK : ORCA
* MILKYWAY_DATABASE :
* LEF_FILE: %LEF%
* TOP_DEF_FILE: %Def%
* LEF_USE_OBS: YES
* METAL_FILL_GDS_FILE :              %DMGDS:i,# Dummy metal GDS file%
* GDS_LAYER_MAP_FILE :               %GdsMap%
* METAL_FILL_POLYGON_HANDLING:       FLOATING

**************
* Extraction *
**************
  MAPPING_FILE : ../MAP/ORCA.map
  TCAD_GRD_FILE : ../NXTGRD/saed32nm_1p9m_Cmin.nxtgrd
*  MODE: 400
  COUPLE_TO_GROUND : NO
  EXTRACT_VIA_CAPS: YES
  INTRANET_CAPS:    NO
  REDUCTION: YES
  REDUCTION_MAX_DELAY_ERROR:         5.0e-13
*  SENSITIVITY: YES
  TEMPERATURE_SENSITIVITY: YES
  OPERATING_TEMPERATURE : -40
```

Extract the .spef files based on the NDM Database & Block Name

Extraction using layer.map file & nxtgrd file

Operation Temperature is -40 in MIN_bst

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STAR RC : Introduction to star.cmd & :run

- Review and check the variables in each scripts
  - **Review $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MIN_bst/star.cmd**

```
**************
* Processing *
**************
*  NUM_PARTS: 2
* REMOVE_FLOATING_NETS: YES
* REMOVE_DANGLING_NETS: YES
* SKIP_CELLS: !
* SKIP_CELLS_FILE:

**************
* Netlisting *
**************
* NETLIST_NAME_MAP: NO
  NETLIST_FILE : ../CAPnNETLIST/ORCA_TOP_MIN_n40c.spef.gz
  NETLIST_FORMAT : SPEF
*  NETLIST_POSTPROCESS_COMMAND: gzip -f

*********
* Noise *
*********
* COUPLING_REPORT_FILE:               coupling.rpt
* COUPLING_REPORT_NUMBER:             1000000
*  COUPLING_ABS_THRESHOLD:             1e-15
*  COUPLING_REL_THRESHOLD:             0.02
```

Path from which the .spef file is extracted & .spef.gz file name

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STAR RC : Introduction to star.cmd & :run

- Review and check the variables in each scripts
  - **Review $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MIN_bst/:run**

```
#module load starrc/2021.06-SP5-5
#StarXtract -clean star.cmd&
#sleep 60
StarXtract star.cmd
```

Run STAR RC based on star.cmd file

  - **Review $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MIN_wst/star.cmd**

```
************
* Database *
************
NDM_DATABASE : ../../rundir/07_chip_finish/nlib/ORCA_chip_finish.nlib
BLOCK : ORCA
* MILKYWAY_DATABASE :
* LEF_FILE: %LEF%
* TOP_DEF_FILE: %Def%
* LEF_USE_OBS: YES
* METAL_FILL_GDS_FILE :                %DMGDS:i,# Dummy metal GDS file%
* GDS_LAYER_MAP_FILE :                 %GdsMap%
* METAL_FILL_POLYGON_HANDLING:         FLOATING

**************
* Extraction *
**************
  MAPPING_FILE : ../MAP/ORCA.map
  TCAD_GRD_FILE : ../NXTGRD/saed32nm_1p9m_Cmin.nxtgrd
*  MODE: 400
  COUPLE_TO_GROUND : NO
  EXTRACT_VIA_CAPS: YES
  INTRANET_CAPS:    NO
  REDUCTION: YES
  REDUCTION_MAX_DELAY_ERROR:           5.0e-13
*  SENSITIVITY: YES
  TEMPERATURE_SENSITIVITY: YES
  OPERATING_TEMPERATURE : 125
```

Extract the .spef files based on the NDM Database & Block Name

Extraction using layer.map file & nxtgrd file

Operation Temperature is 125 in MIN_wst

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STAR RC : Introduction to star.cmd & :run

- Review and check the variables in each scripts
  - **Review $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MIN_wst/star.cmd**

```
**************
* Processing *
**************
*  NUM_PARTS: 2
* REMOVE_FLOATING_NETS: YES
* REMOVE_DANGLING_NETS: YES
* SKIP_CELLS: !
* SKIP_CELLS_FILE:

**************
* Netlisting *
**************
* NETLIST_NAME_MAP: NO
  NETLIST_FILE : ../CAPnNETLIST/ORCA_TOP_MIN_125c.spef.gz
  NETLIST_FORMAT : SPEF
*  NETLIST_POSTPROCESS_COMMAND: gzip -f

*********
* Noise *
*********
* COUPLING_REPORT_FILE:               coupling.rpt
* COUPLING_REPORT_NUMBER:             1000000
*  COUPLING_ABS_THRESHOLD:             1e-15
*  COUPLING_REL_THRESHOLD:             0.02
```

Path from which the .spef file is extracted & .spef.gz file name

  - **Review $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MAX_wst/:run**

```
#module load starrc/2021.06-SP5-5
#StarXtract -clean star.cmd&
#sleep 60
StarXtract star.cmd
```

Run STAR RC based on star.cmd file

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STAR RC: Use ICC2 (IC Compiler II tool) MAX

1. **Your rundir> source $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MAX_bst/:run**
   - **Run STAR RC to source *":run"* file**

```
 < user@linux MAX >> source :run

                                StarRC (TM)

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ExecName: /global/apps/starrc_2021.06-SP5-5/linux64_starrc/bin/StarXtract
Version: S-2021.06-SP5-5
Built on: Aug  6 2023 03:23:19
Start Time: Tue Nov 21 14:50:11 2023
```

```
Block ........... ORCA
Star Directory .. /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/STAR_RC/MAX_bst/star
WARNING: Running StarRC in single corner mode will result in degraded runtime for design flows where
         multiple corners are used.  For optimal runtime, add the necessary commands to configure
         the Simultaneous Multicorner flow (CORNERS_FILE and SELECTED_CORNERS). (SX-3552)
```


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STAR RC: Use ICC2 (IC Compiler II tool) MAX

1. **Your rundir> source $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MAX_bst/:run**
   - **Run STAR RC to source *":run"* file**

```
NDM DB .......
 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/07_chip_finish/nlib/ORCA_chip_finish.nlib:ORCA_chip_finish.nlib

 Ref Libs .......
 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/EXPLORE_physical_only.ndm:EXPLORE_physical_only


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32hvt.ndm:saed32hvt


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32io_fc.ndm:saed32io_fc


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32lvt.ndm:saed32lvt


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32pll.ndm:saed32pll


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32rvt.ndm:saed32rvt


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32sram.ndm:saed32sram
```



```
Netlist_part1        Elp=00:00:04 Cpu=00:00:02 Usr=2.1    Sys=0.4    Mem=396.9
Netlist_assembly1    Elp=00:00:02 Cpu=00:00:01 Usr=1.2    Sys=0.3    Mem=394.8
Netlist_merge_begin  Elp=00:00:00 Cpu=00:00:00 Usr=0.0    Sys=0.0    Mem=330.6
Netlist_merge_end    Elp=00:00:00 Cpu=00:00:00 Usr=0.0    Sys=0.0    Mem=330.6

Done          Elp=00:01:13 Cpu=00:00:45 Usr=40.8   Sys=4.7    Mem=911.3
```


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STAR RC: Use ICC2 (IC Compiler II tool) MAX

1. **Your rundir> source $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MAX_bst/:run**
   - **Check .spef file after extracted (ORCA_TOP_MAX_n40c.spef.gz)**

```
  < user@linux CAPnNETLIST >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/STAR_RC/CAPnNETLIST
  < user@linux CAPnNETLIST >> ls
ORCA_MAX_n40c.spef.gz
```

   - **Open & check *"ORCA_MAX_n40.spef.gz"* file**

```
*SPEF "IEEE 1481-1999"
*DESIGN "ORCA"
*DATE "Tue Nov 21 14:52:06 2023"
*VENDOR "Synopsys"
*PROGRAM "StarRC"
*VERSION "S-2021.06-SP5-5"
*DESIGN_FLOW "PIN_CAP NONE" "NAME_SCOPE LOCAL" "TEMPERATURE 25"
*DIVIDER /
*DELIMITER :
*BUS_DELIMITER []
*T_UNIT 1.00000 NS
*C_UNIT 1.00000 FF
*R_UNIT 1.00000 OHM
*L_UNIT 1.00000 HENRY

// galaxy_comments_begin
//    local_nodeid
// galaxy_comments_end

// COMMENTS
```

```
// GLOBAL_TEMPERATURE 25
//
// TCAD_GRD_FILE ../../NXTGRD/saed32nm_1p9m_Cmax.nxtgrd
//   TCAD_TIME_STAMP Fri Mar 18 15:30:12 2011
//   GRDGENXO_VERSION E-2010.12-SP1-1
//   TCADGRD_VERSION 72:
// TCAD_GRD_FILE ../../NXTGRD/saed32nm_1p9m_Cmax.nxtgrd
//   TCAD_TIME_STAMP Fri Mar 18 15:30:12 2011
//   GRDGENXO_VERSION E-2010.12-SP1-1
//   TCADGRD_VERSION 72:
// TCAD_GRD_FILE ../../NXTGRD/saed32nm_1p9m_Cmax.nxtgrd
//   TCAD_TIME_STAMP Fri Mar 18 15:30:12 2011
//   GRDGENXO_VERSION E-2010.12-SP1-1
//   TCADGRD_VERSION 72:
// TCAD_GRD_FILE ../../NXTGRD/saed32nm_1p9m_Cmax.nxtgrd
//   TCAD_TIME_STAMP Fri Mar 18 15:30:12 2011
//   GRDGENXO_VERSION E-2010.12-SP1-1
//   TCADGRD_VERSION 72
```

Check .spef file refer to 11p~15p

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STAR RC: Use ICC2 (IC Compiler II tool) MAX

1. **Your rundir> source $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MAX_wst/:run**
   - **Run STAR RC to source *":run"* file**

```
 < user@linux MAX >> source :run

                                StarRC (TM)

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           Inclusivity and Diversity" (Refer to article 000036315 at
                       https://solvnetplus.synopsys.com)

ExecName: /global/apps/starrc_2021.06-SP5-5/linux64_starrc/bin/StarXtract
Version: S-2021.06-SP5-5
Built on: Aug  6 2023 03:23:19
Start Time: Tue Nov 21 14:50:11 2023
```

```
Block ........... ORCA
Star Directory .. /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/STAR_RC/MAX_wst/star
WARNING: Running StarRC in single corner mode will result in degraded runtime for design flows where
         multiple corners are used.  For optimal runtime, add the necessary commands to configure
         the Simultaneous Multicorner flow (CORNERS_FILE and SELECTED_CORNERS). (SX-3552)
```


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STAR RC: Use ICC2 (IC Compiler II tool) MAX

1. **Your rundir> source $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MAX_wst/:run**
   - **Run STAR RC to source *":run"* file**

```
NDM DB .......
 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/07_chip_finish/nlib/ORCA_chip_finish.nlib:ORCA_chip_finish.nlib

 Ref Libs .......
 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/EXPLORE_physical_only.ndm:EXPLORE_physical_only


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32hvt.ndm:saed32hvt


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32io_fc.ndm:saed32io_fc


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32lvt.ndm:saed32lvt


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32pll.ndm:saed32pll


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32rvt.ndm:saed32rvt


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32sram.ndm:saed32sram
```



```
Netlist_part1        Elp=00:00:04 Cpu=00:00:02 Usr=2.1    Sys=0.4    Mem=396.9
Netlist_assembly1    Elp=00:00:02 Cpu=00:00:01 Usr=1.2    Sys=0.3    Mem=394.8
Netlist_merge_begin  Elp=00:00:00 Cpu=00:00:00 Usr=0.0    Sys=0.0    Mem=330.6
Netlist_merge_end    Elp=00:00:00 Cpu=00:00:00 Usr=0.0    Sys=0.0    Mem=330.6

Done          Elp=00:01:13 Cpu=00:00:45 Usr=40.8   Sys=4.7    Mem=911.3
```


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STAR RC: Use ICC2 (IC Compiler II tool) MAX

1. **Your rundir> source $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MAX_wst/:run**
   - **Check .spef file after extracted (ORCA_TOP_MAX_125c.spef.gz)**

```
  < user@linux CAPnNETLIST >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/STAR_RC/CAPnNETLIST
  < user@linux CAPnNETLIST >> ls
ORCA_MAX_125c.spef.gz  ORCA_MAX_n40c.spef.gz
```

   - **Open & check *"ORCA_MAX_n40.spef.gz"* file**

```
*SPEF "IEEE 1481-1999"
*DESIGN "ORCA"
*DATE "Thu Dec  7 14:15:11 2023"
*VENDOR "Synopsys"
*PROGRAM "StarRC"
*VERSION "S-2021.06-SP5-5"
*DESIGN_FLOW "PIN_CAP NONE" "NAME_SCOPE LOCAL" "TEMPERATURE 25"
*DIVIDER /
*DELIMITER :
*BUS_DELIMITER []
*T_UNIT 1.00000 NS
*C_UNIT 1.00000 FF
*R_UNIT 1.00000 OHM
*L_UNIT 1.00000 HENRY

// galaxy_comments_begin
//    local_nodeid
// galaxy_comments_end

// COMMENTS
```

```
// GLOBAL_TEMPERATURE 25
//
// TCAD_GRD_FILE ../../NXTGRD/saed32nm_1p9m_Cmax.nxtgrd
//   TCAD_TIME_STAMP Fri Mar 18 15:30:12 2011
//   GRDGENXO_VERSION E-2010.12-SP1-1
//   TCADGRD_VERSION 72:
// TCAD_GRD_FILE ../../NXTGRD/saed32nm_1p9m_Cmax.nxtgrd
//   TCAD_TIME_STAMP Fri Mar 18 15:30:12 2011
//   GRDGENXO_VERSION E-2010.12-SP1-1
//   TCADGRD_VERSION 72:
// TCAD_GRD_FILE ../../NXTGRD/saed32nm_1p9m_Cmax.nxtgrd
//   TCAD_TIME_STAMP Fri Mar 18 15:30:12 2011
//   GRDGENXO_VERSION E-2010.12-SP1-1
//   TCADGRD_VERSION 72:
// TCAD_GRD_FILE ../../NXTGRD/saed32nm_1p9m_Cmax.nxtgrd
//   TCAD_TIME_STAMP Fri Mar 18 15:30:12 2011
//   GRDGENXO_VERSION E-2010.12-SP1-1
//   TCADGRD_VERSION 72
```

Check .spef file refer to 11p~15p

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STAR RC: Use ICC2 (IC Compiler II tool) MIN

1. **Your rundir> source $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MIN_bst/:run**
   - **Run STAR RC to source *":run"* file**

```
 < user@linux MIN >> source :run

                                StarRC (TM)

            Version S-2021.06-SP5-5 for linux64 - Aug 06, 2023

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ExecName: /global/apps/starrc_2021.06-SP5-5/linux64_starrc/bin/StarXtract
Version: S-2021.06-SP5-5
Built on: Aug  6 2023 03:23:19
Start Time: Tue Nov 21 15:14:34 2023
```

```
Block ........... ORCA
Star Directory .. /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/STAR_RC/MIN_bst/star
WARNING: Running StarRC in single corner mode will result in degraded runtime for design flows where
         multiple corners are used.  For optimal runtime, add the necessary commands to configure
         the Simultaneous Multicorner flow (CORNERS_FILE and SELECTED_CORNERS). (SX-3552)
```


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STAR RC: Use ICC2 (IC Compiler II tool) MIN

1. **Your rundir> source $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MIN_bst/:run**
   - **Run STAR RC to source *":run"* file**

```
NDM DB .......
 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/07_chip_finish/nlib/ORCA_chip_finish.nlib:ORCA_chip_finish.nlib

 Ref Libs .......
 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/EXPLORE_physical_only.ndm:EXPLORE_physical_only


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32hvt.ndm:saed32hvt


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32io_fc.ndm:saed32io_fc


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32lvt.ndm:saed32lvt


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32pll.ndm:saed32pll


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32rvt.ndm:saed32rvt


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32sram.ndm:saed32sram
```



```
Netlist_part1        Elp=00:00:04 Cpu=00:00:02 Usr=2.1    Sys=0.4    Mem=396.9
Netlist_assembly1    Elp=00:00:02 Cpu=00:00:01 Usr=1.2    Sys=0.3    Mem=394.8
Netlist_merge_begin  Elp=00:00:00 Cpu=00:00:00 Usr=0.0    Sys=0.0    Mem=330.6
Netlist_merge_end    Elp=00:00:00 Cpu=00:00:00 Usr=0.0    Sys=0.0    Mem=330.6

Done          Elp=00:01:13 Cpu=00:00:45 Usr=40.8   Sys=4.7    Mem=911.3
```


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STAR RC: Use ICC2 (IC Compiler II tool) MIN

1. **Your rundir> source $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MIN_bst/:run**
   - **Check .spef file after extracted (ORCA_TOP_MIN_n40c.spef.gz)**

```
  < user@linux CAPnNETLIST >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/STAR_RC/CAPnNETLIST
  < user@linux CAPnNETLIST >> ls
ORCA_MAX_125c.spef.gz  ORCA_MAX_n40c.spef.gz  ORCA_TOP_MIN_n40c.spef.gz
```

   - **Open & check *"ORCA_MIN.spef.gz"* file**

```
*SPEF "IEEE 1481-1999"
*DESIGN "ORCA"
*DATE "Thu Dec  7 14:29:07 2023"
*VENDOR "Synopsys"
*PROGRAM "StarRC"
*VERSION "S-2021.06-SP5-5"
*DESIGN_FLOW "PIN_CAP NONE" "NAME_SCOPE LOCAL" "TEMPERATURE 25"
*DIVIDER /
*DELIMITER :
*BUS_DELIMITER []
*T_UNIT 1.00000 NS
*C_UNIT 1.00000 FF
*R_UNIT 1.00000 OHM
*L_UNIT 1.00000 HENRY

// galaxy_comments_begin
//    local_nodeid
// galaxy_comments_end

// COMMENTS
```

```
// GLOBAL_TEMPERATURE 25
//
// TCAD_GRD_FILE ../../NXTGRD/saed32nm_1p9m_Cmin.nxtgrd
//   TCAD_TIME_STAMP Fri Mar 18 16:37:38 2011
//   GRDGENXO_VERSION E-2010.12-SP1-1
//   TCADGRD_VERSION 72:
// TCAD_GRD_FILE ../../NXTGRD/saed32nm_1p9m_Cmin.nxtgrd
//   TCAD_TIME_STAMP Fri Mar 18 16:37:38 2011
//   GRDGENXO_VERSION E-2010.12-SP1-1
//   TCADGRD_VERSION 72:
// TCAD_GRD_FILE ../../NXTGRD/saed32nm_1p9m_Cmin.nxtgrd
//   TCAD_TIME_STAMP Fri Mar 18 16:37:38 2011
//   GRDGENXO_VERSION E-2010.12-SP1-1
//   TCADGRD_VERSION 72:
// TCAD_GRD_FILE ../../NXTGRD/saed32nm_1p9m_Cmin.nxtgrd
//   TCAD_TIME_STAMP Fri Mar 18 16:37:38 2011
//   GRDGENXO_VERSION E-2010.12-SP1-1
//   TCADGRD_VERSION 72
```

Check .spef file refer to 11p~15p

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STAR RC: Use ICC2 (IC Compiler II tool) MIN

1. **Your rundir> source $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MIN_wst/:run**
   - **Run STAR RC to source *":run"* file**

```
 < user@linux MIN >> source :run

                                StarRC (TM)

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                       https://solvnetplus.synopsys.com)

ExecName: /global/apps/starrc_2021.06-SP5-5/linux64_starrc/bin/StarXtract
Version: S-2021.06-SP5-5
Built on: Aug  6 2023 03:23:19
Start Time: Tue Nov 21 15:14:34 2023
```

```
Block ........... ORCA
Star Directory .. /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/STAR_RC/MIN_wst/star
WARNING: Running StarRC in single corner mode will result in degraded runtime for design flows where
         multiple corners are used.  For optimal runtime, add the necessary commands to configure
         the Simultaneous Multicorner flow (CORNERS_FILE and SELECTED_CORNERS). (SX-3552)
```


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STAR RC: Use ICC2 (IC Compiler II tool) MIN

1. **Your rundir> source $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MIN_wst/:run**
   - **Run STAR RC to source *":run"* file**

```
NDM DB .......
 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/07_chip_finish/nlib/ORCA_chip_finish.nlib:ORCA_chip_finish.nlib

 Ref Libs .......
 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/EXPLORE_physical_only.ndm:EXPLORE_physical_only


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32hvt.ndm:saed32hvt


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32io_fc.ndm:saed32io_fc


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32lvt.ndm:saed32lvt


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32pll.ndm:saed32pll


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32rvt.ndm:saed32rvt


 /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_ndm/ndm/saed32sram.ndm:saed32sram
```



```
Netlist_part1        Elp=00:00:04 Cpu=00:00:02 Usr=2.1    Sys=0.4    Mem=396.9
Netlist_assembly1    Elp=00:00:02 Cpu=00:00:01 Usr=1.2    Sys=0.3    Mem=394.8
Netlist_merge_begin  Elp=00:00:00 Cpu=00:00:00 Usr=0.0    Sys=0.0    Mem=330.6
Netlist_merge_end    Elp=00:00:00 Cpu=00:00:00 Usr=0.0    Sys=0.0    Mem=330.6

Done          Elp=00:01:13 Cpu=00:00:45 Usr=40.8   Sys=4.7    Mem=911.3
```


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STAR RC: Use ICC2 (IC Compiler II tool) MIN

1. **Your rundir> source $WORK_DIR/2_BACKEND/1_pnr/rundir/STAR/MIN_wst/:run**
   - **Check .spef file after extracted (ORCA_TOP_MIN_125c.spef.gz)**

```
  < user@linux CAPnNETLIST >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/STAR_RC/CAPnNETLIST
  < user@linux CAPnNETLIST >> ls
ORCA_MAX_125c.spef.gz  ORCA_MAX_n40c.spef.gz  ORCA_TOP_MIN_125c.spef.gz  ORCA_TOP_MIN_n40c.spef.gz
```

   - **Open & check *"ORCA_MIN.spef.gz"* file**

```
*SPEF "IEEE 1481-1999"
*DESIGN "ORCA"
*DATE "Thu Dec  7 14:29:07 2023"
*VENDOR "Synopsys"
*PROGRAM "StarRC"
*VERSION "S-2021.06-SP5-5"
*DESIGN_FLOW "PIN_CAP NONE" "NAME_SCOPE LOCAL" "TEMPERATURE 25"
*DIVIDER /
*DELIMITER :
*BUS_DELIMITER []
*T_UNIT 1.00000 NS
*C_UNIT 1.00000 FF
*R_UNIT 1.00000 OHM
*L_UNIT 1.00000 HENRY

// galaxy_comments_begin
//    local_nodeid
// galaxy_comments_end

// COMMENTS
```

```
// GLOBAL_TEMPERATURE 25
//
// TCAD_GRD_FILE ../../NXTGRD/saed32nm_1p9m_Cmin.nxtgrd
//   TCAD_TIME_STAMP Fri Mar 18 16:37:38 2011
//   GRDGENXO_VERSION E-2010.12-SP1-1
//   TCADGRD_VERSION 72:
// TCAD_GRD_FILE ../../NXTGRD/saed32nm_1p9m_Cmin.nxtgrd
//   TCAD_TIME_STAMP Fri Mar 18 16:37:38 2011
//   GRDGENXO_VERSION E-2010.12-SP1-1
//   TCADGRD_VERSION 72:
// TCAD_GRD_FILE ../../NXTGRD/saed32nm_1p9m_Cmin.nxtgrd
//   TCAD_TIME_STAMP Fri Mar 18 16:37:38 2011
//   GRDGENXO_VERSION E-2010.12-SP1-1
//   TCADGRD_VERSION 72:
// TCAD_GRD_FILE ../../NXTGRD/saed32nm_1p9m_Cmin.nxtgrd
//   TCAD_TIME_STAMP Fri Mar 18 16:37:38 2011
//   GRDGENXO_VERSION E-2010.12-SP1-1
//   TCADGRD_VERSION 72
```

Check .spef file refer to 11p~15p

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Thank You

